MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION, MUMBAI

A Project Report on

Songir Handcraft Utensil Portal

Submitted in partial fulfillment for the award of

Diploma in Computer Engineering

Submitted by

Kasar Jidnyasa Govind

Pardeshi Nandini Jitendra

Patil Jagruti Chandrakant

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DEPARTMENT OF COMPUTER ENGINEERING

S.S.V.P.S's BapuSaheb Shivajirao Deore Polytechnic, Dhule

(Institute Code : 0059)

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Pardeshi Nandini Jitendra	23610960240
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Acknowledgment

We would like to express our sincere gratitude to everyone who supported and guided us in completing our project on “Songir Handcraft Utensils portal”.

First and foremost, we express our heartfelt thanks to our respected Head of Department, Mr. N. D. Patel , for his constant encouragement and support. His guidance helped us to complete this project successful. We would also like to express our deep gratitude to our project mentor, Ms. Ratna Patil, for her valuable guidance, suggestions, and continuous supervision throughout the project. Her support and motivation played an important role in the successful completion of this work.

We are also thankful to the local artisans and craftsmen of Songir, who shared their knowledge, experience, and valuable information about the traditional handcrafted utensils. Their cooperation helped us understand the cultural and practical importance of this craft.

We would like to extend our thanks to all the faculty members of our department for their support and encouragement during this project.

Finally, we express our sincere thanks to our parents, friends, and all those who directly or indirectly helped us in completing this project successfully.

Your sincerely,

Kasar Jidnyasa Govind

Pardeshi Nandini Jitendra

Patil Jagruti Chandrakant

Sanyasi Tanu Kiran

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Abstract

The Songir Handcrafted Brass and Copper Utensils Online Marketplace is a web-based platform designed to promote and support the traditional utensil artisans of Songir village who are skilled in crafting brass and copper utensils using traditional techniques. The primary objective of this system is to create a digital marketplace where multiple shopkeepers and utensil artisans can showcase and manage their handcrafted products on a single platform.

The platform enables customers to easily explore a wide range of handcrafted utensils, compare products, and place quote requests for customized or bulk orders according to their requirements. and “Add My Shop” functionality , by providing an organized and accessible system, the platform improves communication between local utensil artisans and customers while expanding the market reach of local handcrafted products.

This project aims to digitize the traditional utensil craft industry, provide greater visibility and economic opportunities for utensil artisans, and preserve the cultural heritage associated with brass and copper utensil craftsmanship in Songir. By integrating traditional craftsmanship with modern web technology, the system supports sustainable development and ensures that the valuable skills of local utensil artisans continue to be recognized and appreciated in the modern marketplace.

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CHAPTER 1

INTRODUCTION AND BACKGROUND OF INDUSTRY

1.1 INTRODUCTION

Traditional handicrafts play an important role in preserving cultural heritage and supporting local economies. One such traditional craft is the production of handcrafted brass and copper utensils by the skilled utensil artisans of Songir village. These artisans use traditional techniques and tools to produce durable, high-quality, and culturally significant products.

Handcrafted brass and copper items are widely valued for their durability, traditional design, and health benefits. The artisans of Songir are known for creating a wide range of products including plates, bowls, cooking vessels, decorative utensils, and statues made from brass and copper metals. These products reflect the rich craftsmanship and cultural identity of the region.

However, many utensil artisans face difficulties in reaching a larger market because their products are mainly sold through local shops and small marketplaces. Due to limited digital presence, their handcrafted products often do not receive sufficient visibility and recognition.

To address this challenge, the Songir Handcrafted Brass and Copper Utensils Online Marketplace is proposed as a web-based platform that enables multiple artisans and shopkeepers to showcase their handcrafted products online. The platform allows customers to explore products, compare options, and request customized or bulk orders easily. This system helps promote traditional craftsmanship while connecting utensil artisans with a wider customer base through digital technology.

1.2 BACKGROUND OF INDUSTRY

The brass and copper utensil industry in Songir has been an important part of the village's traditional economy for many years. Local utensil artisans possess specialized skills in shaping, polishing, and crafting various products using brass and copper metals.These artisans produce a variety of handcrafted items such as plates, bowls, cooking vessels, decorative utensils, and statues, which are widely appreciated for their durability, traditional value, and craftsmanship.Each product is carefully handmade using traditional methods that have been passed down through generations.

Despite the quality and uniqueness of these handcrafted products, the industry mainly operates through small-scale shops and local markets. This limits the ability of artisans to reach a broader audience and compete with mass-produced products available in modern markets.

With the rapid growth of digital technology and online marketplaces, there is an opportunity to modernize this traditional industry. An online platform can help utensil artisans promote their handcrafted products, expand their market reach, and improve their economic opportunities while preserving the traditional craftsmanship of Songir.

CHAPTER 2
LITERATURE SURVEY AND SPECIFICATION

2.1 STUDY OF EXISTING SYSTEM / REVIEW OF RESEARCH PAPERS

The study of existing e-commerce systems and online marketplaces shows that many digital platforms provide facilities for product listing, seller registration, and online product promotion. These systems allow multiple sellers to showcase their products on a single platform and enable customers to explore, compare, and purchase products conveniently through the internet.

Research related to handicraft and small-scale industries indicates that digital platforms can significantly improve the visibility and market reach of traditional products. Online marketplaces help artisans promote their products beyond local markets and connect with a larger customer base.

However, most existing platforms are designed mainly for large-scale businesses and commercial products. They usually do not focus on the specific needs of traditional utensil artisans who produce handcrafted items using traditional methods. As a result, many artisans from rural areas are unable to effectively promote their products through modern digital systems.

2.2 LIMITATIONS OF EXISTING SYSTEM / PROBLEMS DISCUSSED IN RESEARCH PAPERS

Although existing online marketplaces provide many useful features, several limitations are observed when it comes to supporting traditional utensil artisans.

Most platforms are designed for large businesses and are not suitable for small artisans who may have limited technical knowledge. These systems are often complex and require advanced management tools, which makes them difficult for rural artisans to use.

Another limitation is the absence of a dedicated platform for handcrafted brass and copper utensils. Customers who want to purchase traditional products often face difficulty in finding authentic handcrafted items and comparing products from different artisans.

In addition, existing systems usually do not provide a simple way for customers to request customized or bulk orders, which is a common requirement in traditional utensil markets

2.3 PROBLEM IDENTIFICATION

The utensil artisans of Songir village produce high-quality handcrafted brass and copper utensils using traditional skills and techniques. However, their products are mostly sold through local markets and physical shops, which limits their ability to reach a wider audience.

Due to the lack of a centralized digital platform, artisans face challenges in promoting their products online and expanding their market reach. Customers also experience difficulty in exploring various products, comparing offerings from different shopkeepers, and placing custom or bulk order requests.

Therefore, there is a need for an online system that allows multiple utensil artisans and shopkeepers to showcase their handcrafted products on a single digital platform, making it easier for customers to explore and interact with them.

2.4 PROBLEM DEFINITION

The proposed system is a **web-based online marketplace** designed to support the promotion and sale of handcrafted brass and copper utensils produced by the utensil artisans of Songir village.

The system will allow artisans and shopkeepers to list and manage their products online, while customers can easily explore different items such as plates, bowls, cooking vessels, decorative utensils, and statues made from brass and copper metals.

The platform will also provide features for product comparison and allow customers to send requests for customized or bulk orders. The system aims to provide a simple, user-friendly interface that helps artisans promote their traditional products and connect with a wider customer base through digital technology

2.5 QUESTIONNAIRE FOR REQUIREMENTS SURVEY

During our literature survey, we interacted with both **local artisans/sellers** and **customers** to understand the current handicraft utensil market, challenges, and user requirements.

Questions for Artisans / Sellers (Handcraft Utensil Providers):

a) What types of handcraft utensils do you produce or sell?

b) What materials are used in making these utensils?

c) How do you currently promote or sell your products?

d) What challenges do you face in reaching more customers?

e) Do you maintain any record of your products and sales?

f) What pricing strategy do you follow for your products?

g) Are you interested in selling your products through an online platform?

h) What kind of support or features would you expect from a digital system?

i) How do you handle orders and payments currently?

j) Do you face any issues related to delivery or logistics?

Questions for Customers (Buyers / Users):

a) How do you usually search for handcraft utensils?

b) What factors do you consider before purchasing a utensil?

c) Do you prefer traditional/local products over factory-made items? Why?

d) What type of utensils do you mostly look for?

e) What price range do you prefer for handcraft products?

f) Do you check reviews or ratings before buying a product?

g) What difficulties do you face while buying handcraft utensils?

h) Would you prefer an online platform to explore and purchase these products?

i) What features would you like in such a system (e.g., product details, reviews, easy payment)?

j) How important is product authenticity and quality to you?

CHAPTER 3

SCOPE OF PROJECT

The scope of this project is to design and develop a dedicated digital marketplace platform for promoting the handcrafted brass and copper utensil industry of Songir village. The system aims to provide a structured digital platform where local artisans and shopkeepers can showcase and sell their handcrafted products beyond traditional offline markets.

The system focuses on developing a multi-vendor online marketplace, allowing multiple artisans and shopkeepers to register, manage, and display their products independently. The platform will enable customers to explore a variety of traditional handcrafted items such as plates, bowls, cooking utensils, decorative items, and statues made from brass and copper.

- **Roles and Functional Scope**

1) Shopkeeper / Artisan Module

- Register and create their profile on the platform
- Add, update, and manage product listings (name, price, description, images)
- Display different categories of handcrafted utensils
- Handle customer inquiries and custom order requests
- Maintain product availability and basic stock information
- View customer interest and manage order requests manually

2) Customer Module

- Browse and search for various handcrafted utensils
- View detailed product information and images
- Compare different products based on price and type
- Send inquiries or request customized/bulk orders
- View basic product ratings or feedback (if available)
- Interact with sellers for more details about products

3) Admin Module

- Manage user accounts (shopkeepers and customers)
- Monitor product listings and platform activities
- Approve or remove inappropriate product content
- Maintain overall system data and ensure smooth functioning
- Handle basic system control and data organization

System Features within Scope

- Simple and user-friendly web interface
- Multi-vendor product listing system
- Product categorization and search functionality
- Vendor–customer interaction through inquiries
- Basic administrative controls

Out of Scope (Future Enhancements)

- Online payment integration
- Automated delivery and logistics management
- Real-time order tracking system
- Advanced analytics and recommendation system

Conclusion

Overall, this project serves as a foundational step toward the digital transformation of the traditional utensil industry in Songir. It helps artisans and shopkeepers expand their market reach while preserving the cultural heritage and craftsmanship of handcrafted brass and copper utensils.

CHAPTER 4

METHODOLOGY

The project follows a **structured Software Development Life Cycle (SDLC)** approach to ensure systematic planning, design, and development of the proposed system.

1. Requirement Identification

The first step involved identifying the requirements of the system by studying the traditional utensil market of Songir village. The needs of **utensil artisans, shopkeepers, and customers** were analyzed to understand the challenges faced in selling handcrafted products through offline markets. This study helped identify the lack of digital presence and the need for a centralized online marketplace.

2. System Planning and Design

Based on the identified requirements, a multi-vendor online marketplace model was planned. The overall system architecture was designed to support multiple shopkeepers managing their product listings on a single platform. The database structure and user roles such as Customer, Shopkeeper, and Administrator were defined clearly. The user interface design was planned to use a copper and brass themed layout to represent the cultural identity of traditional utensil craftsmanship.

3. System Development

During this phase, the structure of the web-based platform was developed. Features such as product listing, product display, shop comparison, and custom or bulk order request functionality were incorporated. The system was designed to provide a simple and user-friendly interface for both artisans and customers.

4. Testing and Evaluation

After development, the system model was evaluated to ensure that all features function correctly. The usability of the platform was analyzed to confirm that artisans can easily manage their products and customers can conveniently explore and interact with the system.

5. Implementation

Finally, the proposed system model demonstrates how the digital marketplace can be implemented to support **Songir's utensil artisans**. The system provides a structured platform for promoting handcrafted brass and copper utensils and helps bridge the gap between traditional craftsmanship and modern digital technology.

4.1 BLOCK DIAGRAM OF SYSTEM

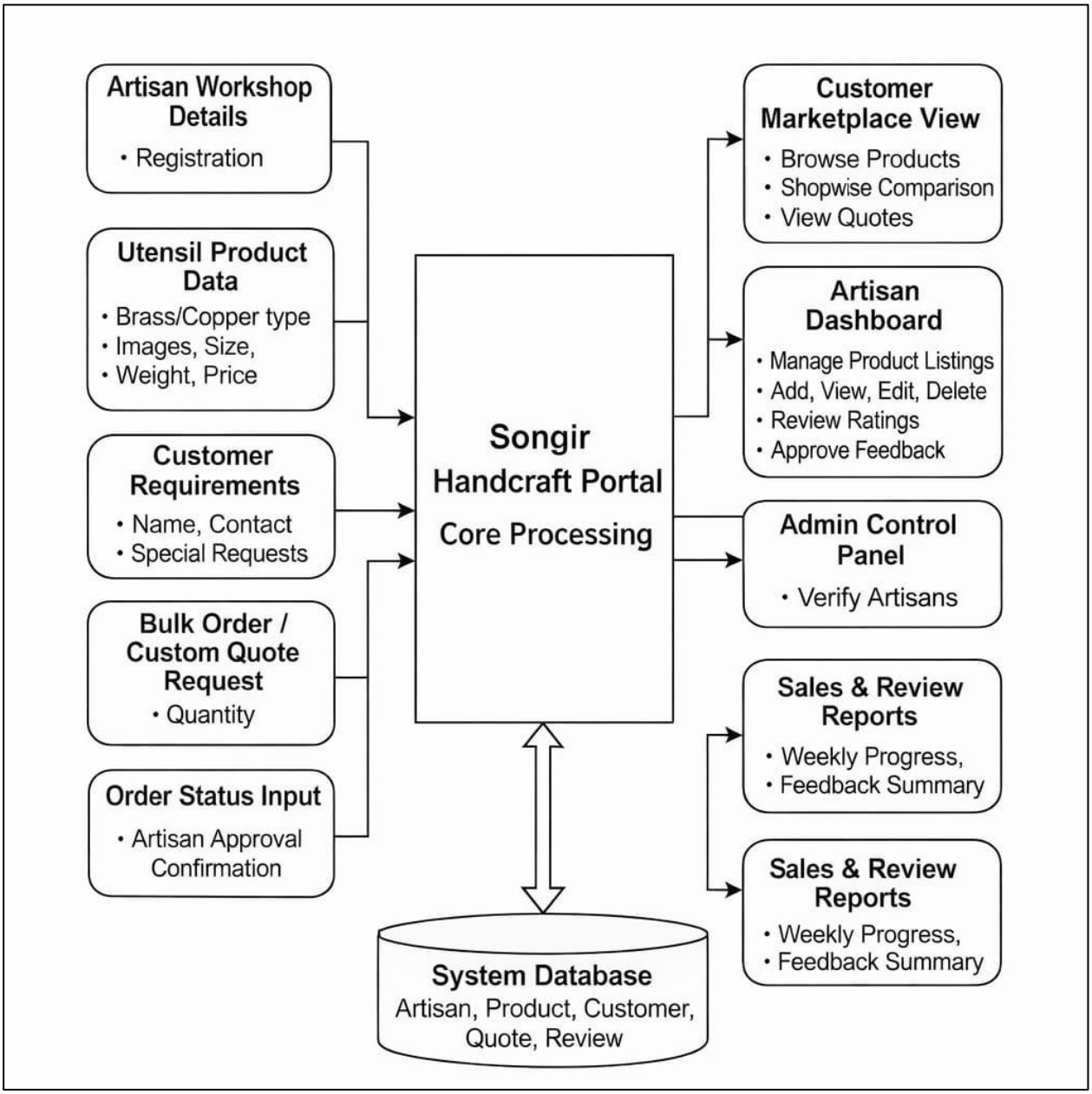

Figure 4.1 Block diagram of Songir Handcraft Utensil Portal

The diagram represents the working of the Songir Handcraft Portal Core Processing System, which connects customers, artisans (shopkeepers), and admin through a centralized platform.

Initially, the process starts with artisan workshop registration, where artisans register their details in the system. Along with this, artisans add utensil product data, including material type (brass/copper), images, size, weight, and price. This information is stored and managed by the system.

On the customer side, users provide their requirements, such as name, contact details, and any special requests. If customers want to place large orders, they can submit a bulk order or custom

quote request by specifying quantity and requirements. These inputs are sent to the core processing system.

The core processing system acts as the central unit that handles all operations. It processes customer requests and forwards them to the respective artisans. Artisans then review the request and provide approval or confirmation, which is updated in the system as order status input.

Customers can access the marketplace view, where they can browse products, compare different shops, and view quotations provided by artisans. This helps customers make better decisions before placing orders.

On the other hand, artisans use their dashboard to manage product listings. They can add, view, edit, or delete products. They can also review customer ratings and feedback and approve feedback where required.

The admin control panel is responsible for verifying artisans and maintaining system authenticity. Admin ensures that only valid artisans are allowed on the platform and monitors overall system activities.

The system also generates sales and review reports, which include weekly progress and feedback summaries. These reports help in analyzing performance and improving services.

All the data, including artisan details, product information, customer data, quotes, and reviews, is stored in the system database, which acts as a central storage and ensures smooth data flow between all modules.

4.2 ACTION PLAN:

The action plan describes the sequence of activities carried out for the successful development of the proposed system.

1.Problem Identification

The traditional brass and copper utensil market of Songir was studied to identify the challenges faced by utensil artisans in promoting and selling their handcrafted products.

2. Requirement Collection

The system requirements were identified by analyzing the needs of customers, utensil artisans, and shopkeepers, focusing on features such as product display, shop registration, and order requests.

3. System Planning

The structure of the online marketplace was planned, including the definition of system modules, user roles, and overall workflow of the platform.

4. System Design

The interface design, database structure, and platform layout were prepared to ensure a simple and organized system.

5. Development of Key Modules

Important modules such as **Add My Shop, product listing, product exploration, and custom order request** were developed for the platform.

6. Testing and Evaluation

The developed system model was tested to ensure proper functionality, usability, and smooth interaction between artisans and customers.

7. Final Documentation

All system processes and project details were documented to demonstrate the working model of the proposed digital marketplace.

CHAPTER 5

DETAILS OF DESIGNS, WORKING AND PROCESSES

5.1 WORKING OF SYSTEM :-

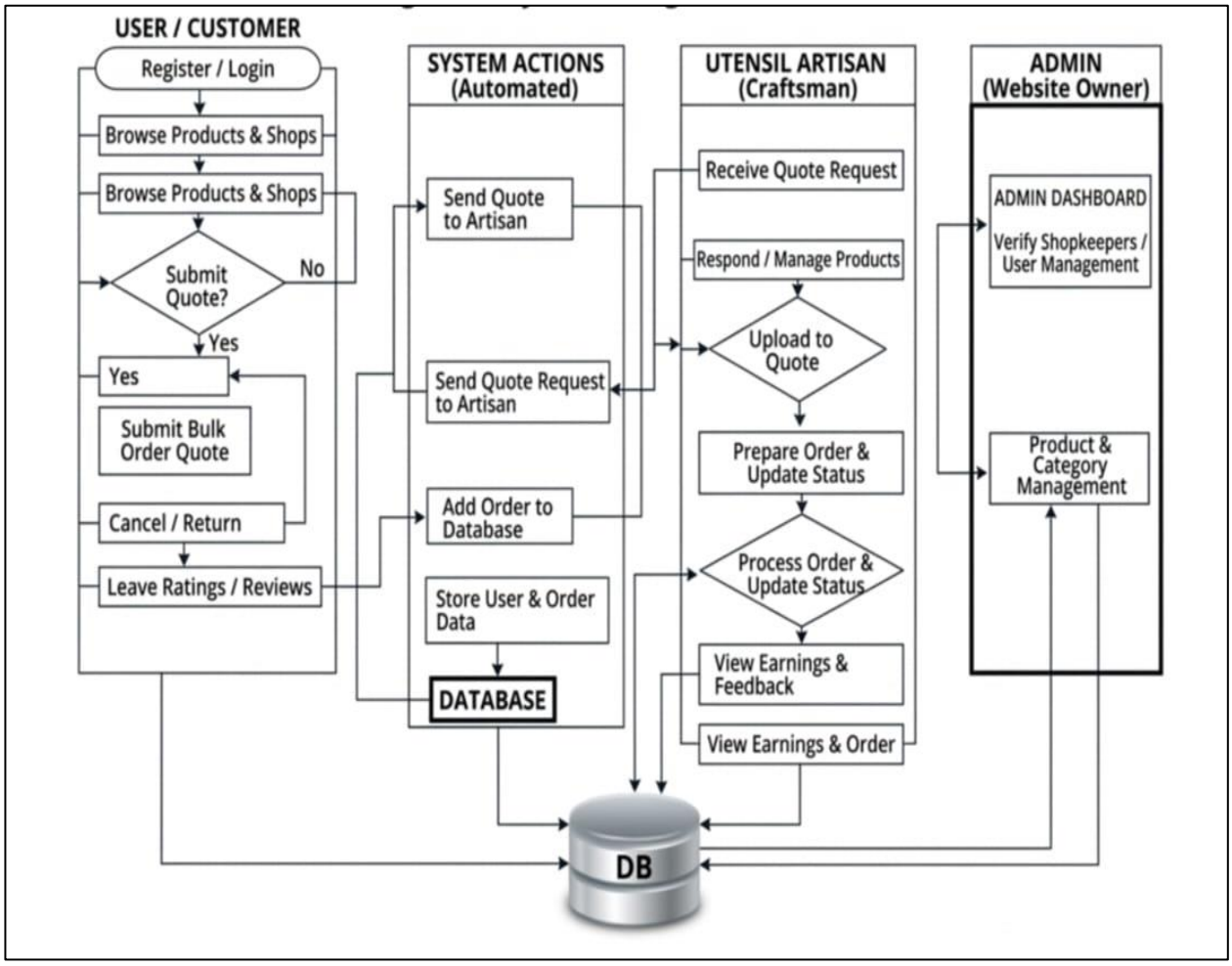

Fig 5.1 Working of Songir Handcraft Utensils Portal

Figure 5.1 shows the flow diagram of the system in which three user roles are displayed: Customer, Shopkeeper, and Admin. Step-wise role of every user in the system is represented.

Initially, shopkeeper registration is done by admin, and the admin verifies shopkeepers before allowing them to access the system. Then the admin manages product categories and overall system setup. After that, customers register into the system through the registration form provided to them.

Once registration is completed, customers can browse products and shops available on the platform. If the customer is interested, they can send a quote request or place a bulk order request. The system forwards this request to the respective shopkeeper.

After receiving the request, the shopkeeper reviews and responds to the quote request, and uploads quotation details. From this stage onward, the role of both customer and shopkeeper becomes important, and it is a continuous process.

The system then follows a process where the shopkeeper prepares the order, updates order status, and processes the order. On the other side, the customer can track order status, manage orders, and perform actions like cancel or return if required. The customer can also add ratings and reviews after completion of the order.

Throughout the process, the system stores all data such as user details, order information, and transaction records in the database. The admin monitors all activities, manages users, and controls product and category details.

At the end, the system maintains complete records of each transaction and user activity, which helps in tracking performance, feedback, and overall system efficiency.

5.2 ADMIN MODULE WORKFLOW

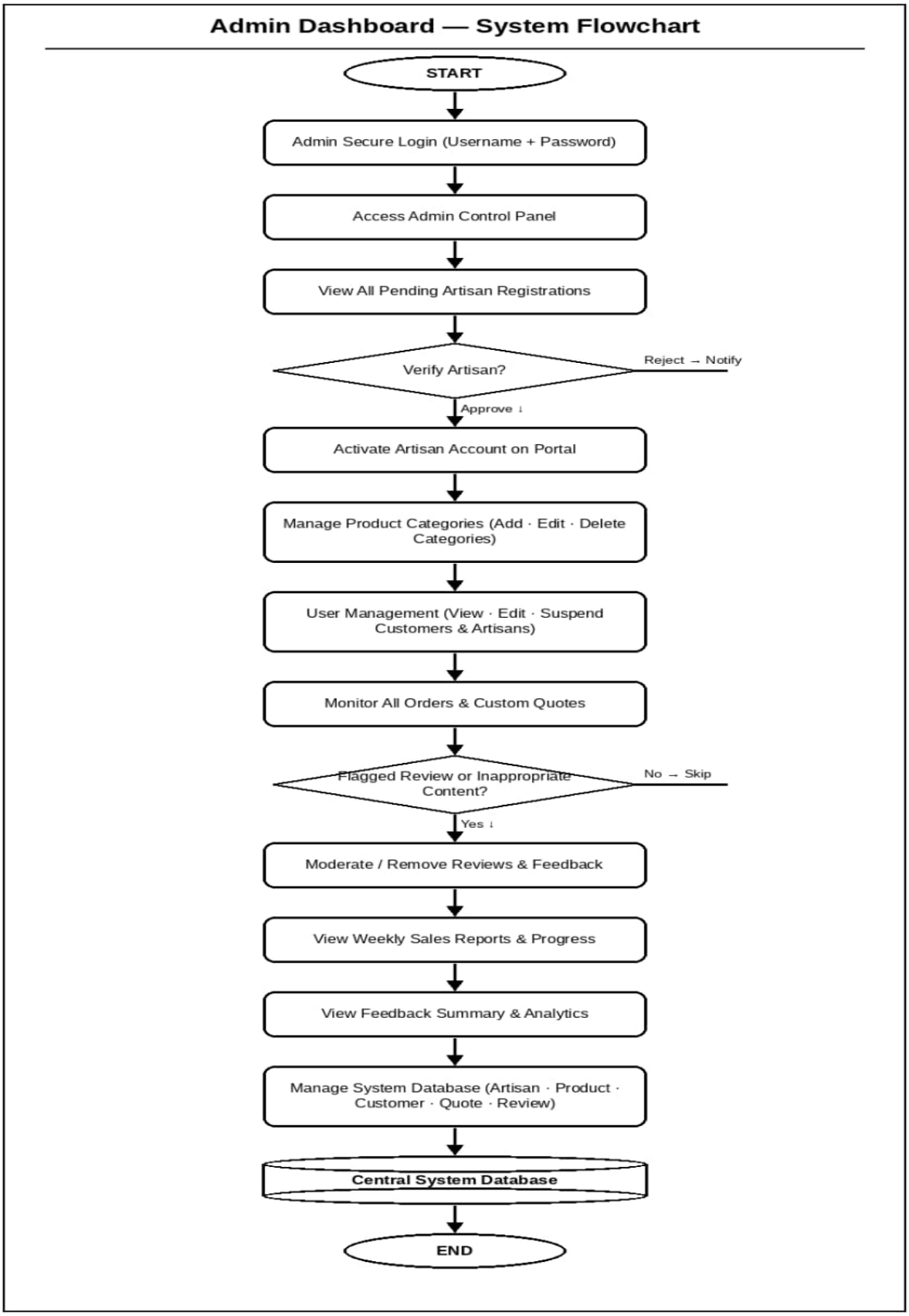

Fig 5.2 Admin Module Flowchart

Admin first logs into the system using secure credentials and accesses the admin control panel. From the dashboard, the admin reviews pending artisan registrations and either approves or rejects them.

After verification, the admin manages product categories by adding, editing, or deleting them. The admin also handles user management, including viewing users and controlling their accounts.

Next, the admin monitors all orders and custom quotes to track system activity. If any review or content is flagged, the admin checks and moderates it.

Finally, the admin views sales reports and feedback analytics and manages the system database to maintain overall system performance.

5.1 Admin Feature table:-

Table 1: Admin Features

Admin Feature	**Description**
Artisan Verification	Approve or reject new artisan registrations
User Management	View, edit, and suspend customers and artisans
Category Management	Add, edit, delete product categories
Order Monitoring	Oversee all orders and custom quotes portal-wide
Content Moderation	Review and remove inappropriate reviews/feedback
Sales Analytics	Weekly reports, earnings summary, performance data
Database Management	Oversee all portal data integrity and records

5.3 SHOPKEEPER MODULE WORKFLOW :-

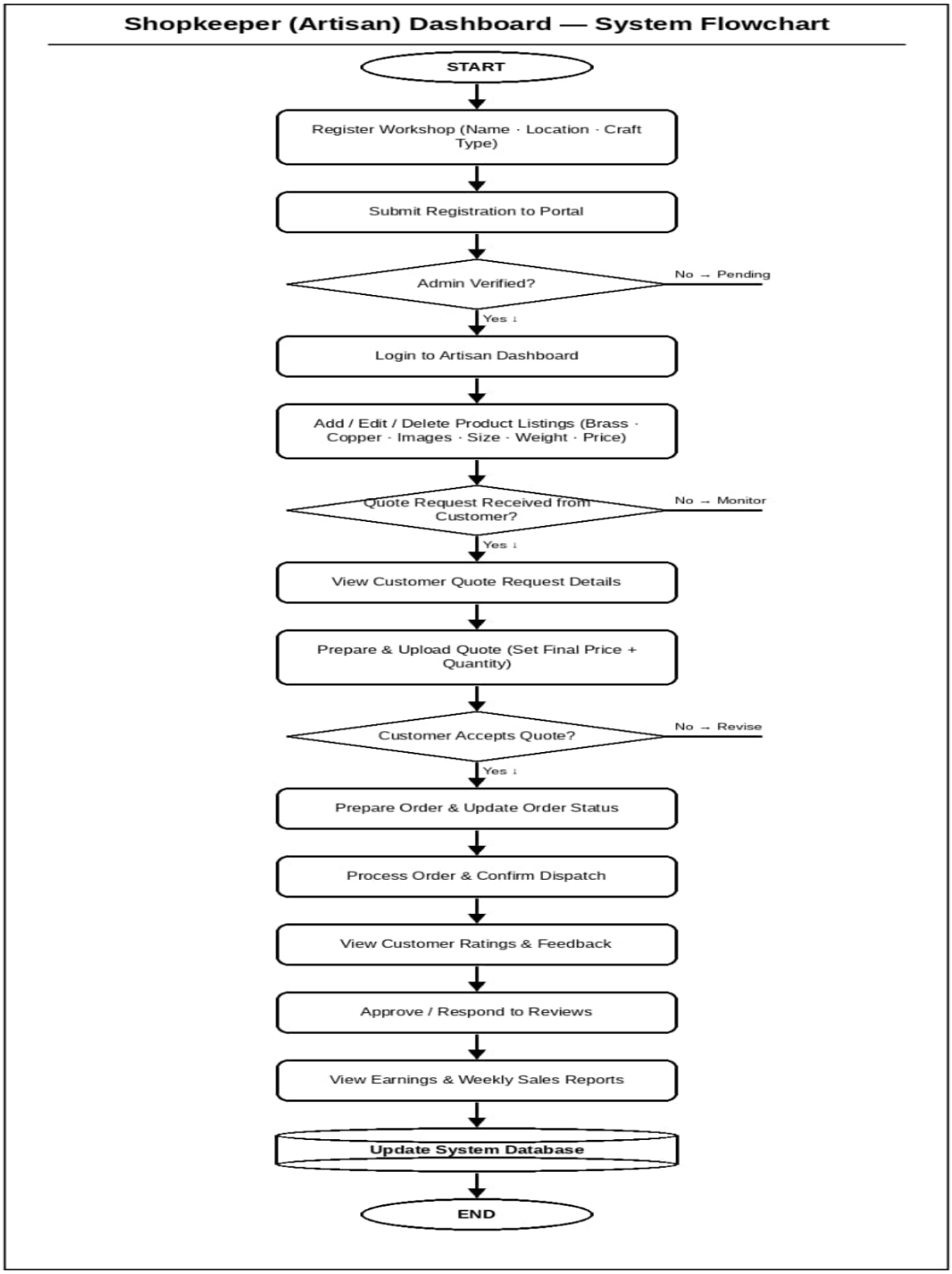

Fig 5.3 Shopkeeper Module Workflow

The shopkeeper first registers their workshop by submitting required details, which are verified by the admin. Once approved, the shopkeeper logs into the dashboard.

After login, the shopkeeper manages product listings by adding, editing, or deleting products. The system then checks for any customer quote requests. If a request is received, the shopkeeper prepares and uploads a quote.

If the customer accepts the quote, the shopkeeper processes the order, updates its status, and confirms dispatch. After delivery, the shopkeeper can view and respond to customer feedback.

Finally, the shopkeeper monitors earnings and reports, while all activities are automatically stored in the system database.

5.2 Shopkeeper Feature:-

Table 2: Shopkeeper Feature

Shopkeeper Feature	**Description**
Product Management	Add, edit, delete products with images and pricing
Quote Management	Receive and respond to customer bulk/custom quotes
Order Processing	Prepare, dispatch and track order status
Review Management	View, respond to, and flag customer reviews
Earnings Report	Weekly sales data, earnings, and performance summary
Dashboard Access	Secure login with verified artisan credentials

5.4 USER WORKFLOW :-

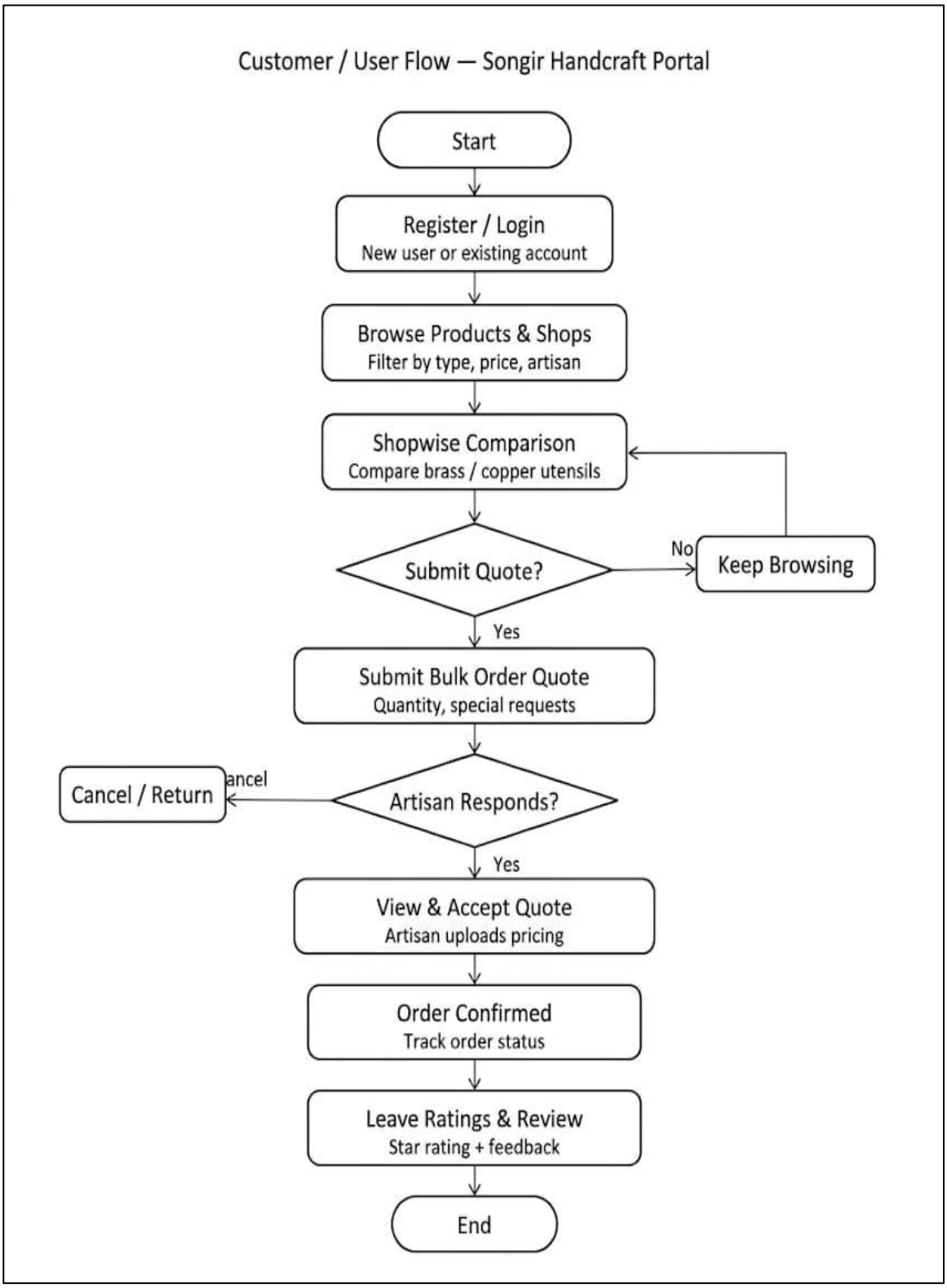

Fig 5.4 User Workflow

The Customer/User Flow diagram illustrates the step-by-step interaction of a user with the Songir Handcraft Portal, starting from accessing the system to completing an order and providing feedback.

Initially, the process begins with the Start point, where the user accesses the portal. The user is then required to either register as a new user or login using existing credentials. This step ensures authentication and secure access to the system.

After successful login, the user can browse products and shops available on the platform. The system provides filtering options such as product type, price range, and artisan details, which help users to easily find relevant products.

Next, the user can make a Shopwise comparison of products, particularly focusing on items like brass or copper utensils. This feature enables the user to evaluate different options and select the most suitable product based on quality and pricing.

At this stage, a decision point is reached where the user chooses whether to submit a quote request. If the user decides not to proceed, they can continue browsing. If the user chooses to proceed, they submit a bulk order quote, specifying the required quantity and any special requirements.

Once the quote is submitted, the system waits for the artisan's response. If the artisan does not respond or the user cancels the request, the process ends or returns to browsing. If the artisan responds, they provide pricing details for the requested order.

The user can then view and accept the quote provided by the artisan. Upon acceptance, the order is confirmed, and the user can track the status of the order through the system.

Finally, after completion of the transaction, the user is given the option to leave ratings and reviews, providing feedback and a star rating based on their experience. This helps improve the overall quality and trustworthiness of the platform.

The process concludes at the End point, indicating the completion of the user interaction cycle.

5.5 USECASE DIAGRAM FOR SONGIR HANDCRAFT UTENSIL PORTAL :-

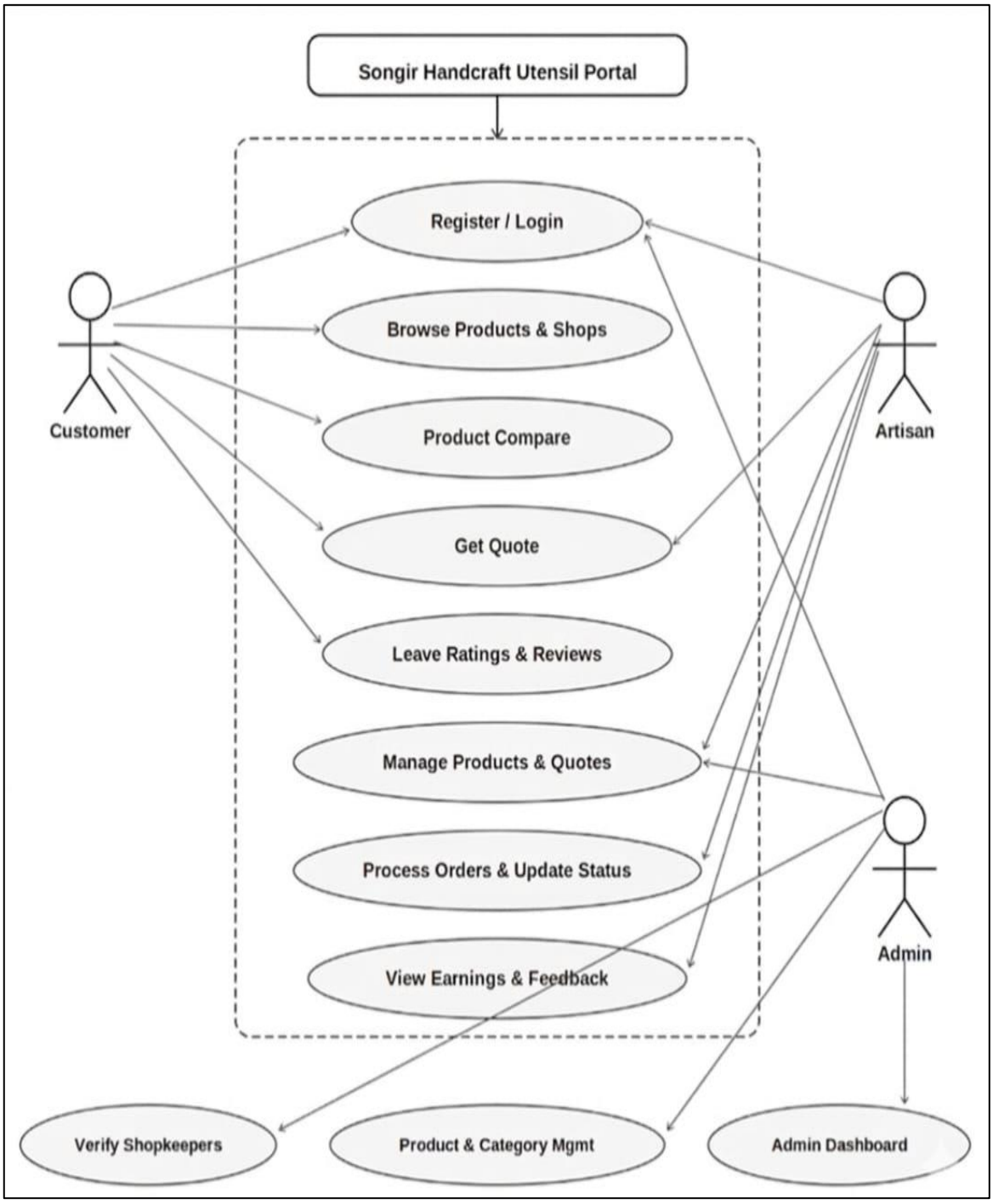

Fig 5.5 Usecase Diagram for Songir Handcraft utensil portal

The **Songir Handcraft Utensil Portal** use case diagram represents an online system designed to connect customers with local artisans who sell handcrafted utensils. The system includes three main actors: Customer, Artisan, and Admin, each having specific roles and interactions within the platform.

The customer first registers or logs into the system and can then browse different products and shops, compare items, request price quotations, and provide ratings and reviews based on their experience. The artisan, who acts as a seller, also registers or logs in to the portal and is responsible for managing products and responding to customer quotations.

They process incoming orders, update the order status, and can view their earnings along with customer feedback. The admin plays a supervisory role by verifying shopkeepers to ensure authenticity, managing product categories and listings, and monitoring the overall system through an admin dashboard.

The complete workflow shows that customers explore and request products, artisans fulfill those requests and manage sales, while the admin ensures smooth operation and control of the platform.

This diagram shows how different users interact with the **Songir Handcraft Utensil Portal**. There are three main users:

- **Customer** – browses products, compares items, gets quotations, and gives reviews.
- **Artisan** – manages products, handles orders, updates status, and views earnings.
- **Admin** – manages categories, verifies shopkeepers, and controls the system.

The system allows users to register/login, explore products, request quotes, and manage orders.

It shows the complete working of the online marketplace and how each user performs their tasks.

CHAPTER 6

RESULT AND APPLICATION

6.1 RESULT:

- **Registration page:**

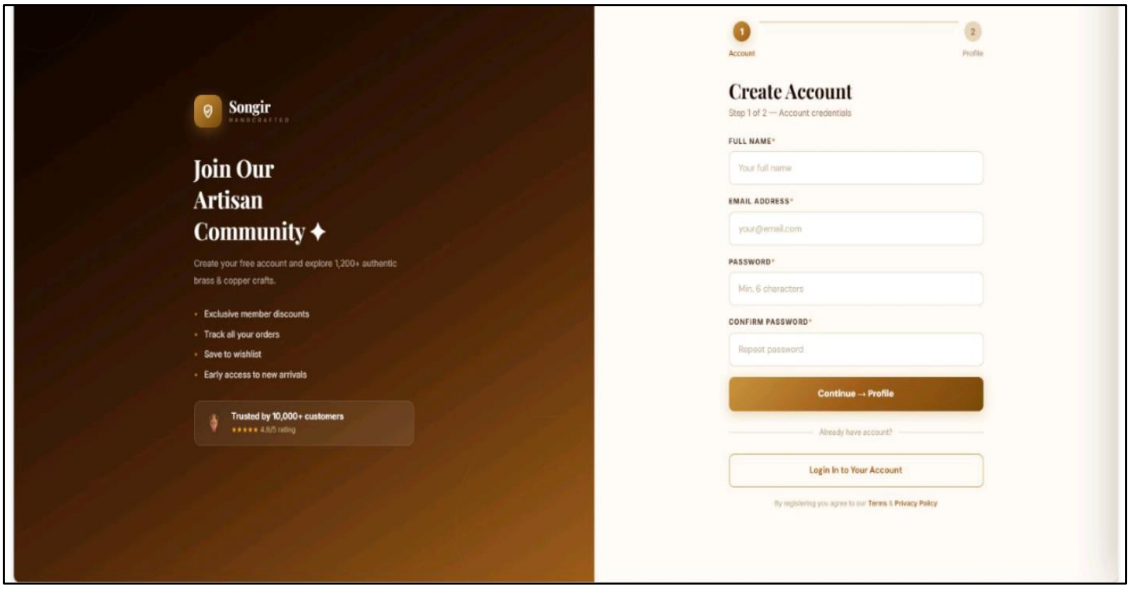

Fig 6.1 Registration page

The Registration page allows new customers to create a free account. It follows a 2-step flow: Step 1 collects Account Credentials (name, email, password, confirm password) and Step 2 collects Profile information (avatar, city, phone). The left panel shows the brand value proposition to encourage sign-ups.

Table 3: Registration page Screen Element-wise Explanation

UI Element / Section	Explanation
Left Panel — Brand Story	Shows Songir logo, 'Join Our Artisan Community' heading, and a short description of the platform's mission.
Left Panel — Benefit Points	4 trust points: Discover Authentic Crafts, Support Local Artisans, Exclusive Deals, Secure Shopping.
Step Indicator	Two-step progress indicator at top of the form — Step 1: Account, Step 2: Profile.
Full Name Field	Text input for the user's complete name.

UI Element / Section	Explanation
Email Address Field	Email input for the new account's login email address.
Password Field	Password input (min. 6 characters required) with visibility toggle.
Confirm Password Field	Re-enter password for validation — client-side check ensures both match.
Continue to Profile Button	Primary button that validates Step 1 data and moves to Step 2 (Profile info).
Step 2 — Avatar Upload	Click-to-upload profile photo option (stored on Cloudinary).
Step 2 — Phone / City	Additional profile fields for phone number and city of residence.
Auto Login on Submit	After final submission, user is automatically logged in and redirected to the Homepage.

Table 4: APIs Called on this Screen

Method	Endpoint	Purpose	Body / Params
POST	/api/auth/register	Register new user	*name, email, password, role*
POST	/api/auth/login	Login, returns JWT token	*email, password*
POST	/api/auth/logout	Logout, sets Inactive status	*userId*
GET	/api/auth/users	Get all users (Admin only)	-

Home page(Landing Page)

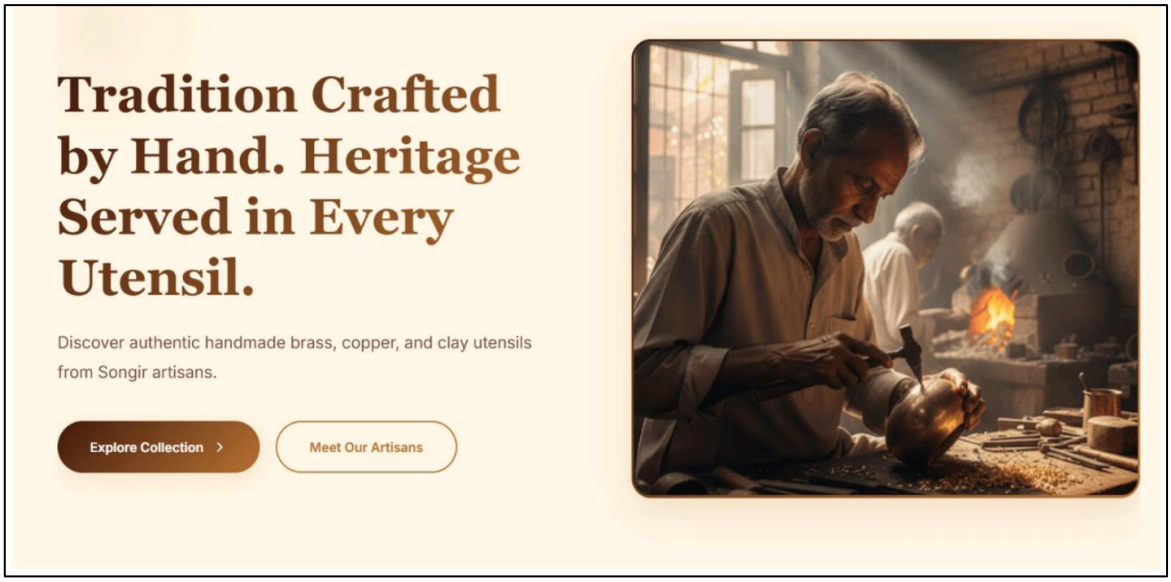

Fig 6.2 Home page(Landing Page)

The Homepage is the main landing page for customers visiting the Songir portal. It showcases the brand story, hero banner, popular products, featured shops, and navigation to all sections. The warm earthy color scheme reflects the traditional craft identity.

Sections on this Page

Top announcement bar: 'New: Copper Masala Dibba & Brass Puja Thali Sets now in stock!'

Navigation: Home, Products, Shops, About, Contact + Search icon, Wishlist, Cart, Registration, Get Quote, +Add Shop

Hero Section: Main headline 'Tradition Crafted by Hand. Heritage Served in Every Utensil.' with CTA buttons

Popular Utensils grid: Featured products with ratings and prices

Shop listings, about section, Contact section below the fold.

Functional Breakdown

Navigation bar fetches cart count and wishlist count from local state/context

'Explore Collection' button navigates to /products page

'Meet Our Artisans' button navigates to /shops page + Add Shop button opens the Shop Registration form

Table 5: Home Page Screen Element-wise Explanation

UI Element	Explanation
Top Announcement Bar	Scrolling banner showing offers - Free shipping on orders above ₹999 and 15% off coupon code SONGIR15.
Navigation Bar	Logo on left, main nav links (Home, Products, Shops, About, Contact), search icon, wishlist, cart, Registration button, Get Quote, and Add Shop buttons on right.
Hero Heading	Bold serif text: 'Tradition Crafted by Hand. Heritage Served in Every Utensil.' - the core brand tagline.
Sub-tagline	Short description: 'Discover authentic handmade brass, copper, and clay utensils from Songir artisans.'
Explore Collection Btn	Dark brown pill button - navigates to the Products listing page.
Meet Our Artisans Btn	Outline pill button - navigates to the Shops listing page.
Hero Image	High-quality image of a real artisan at work in a traditional workshop, building trust with the visitor.

Product Page:

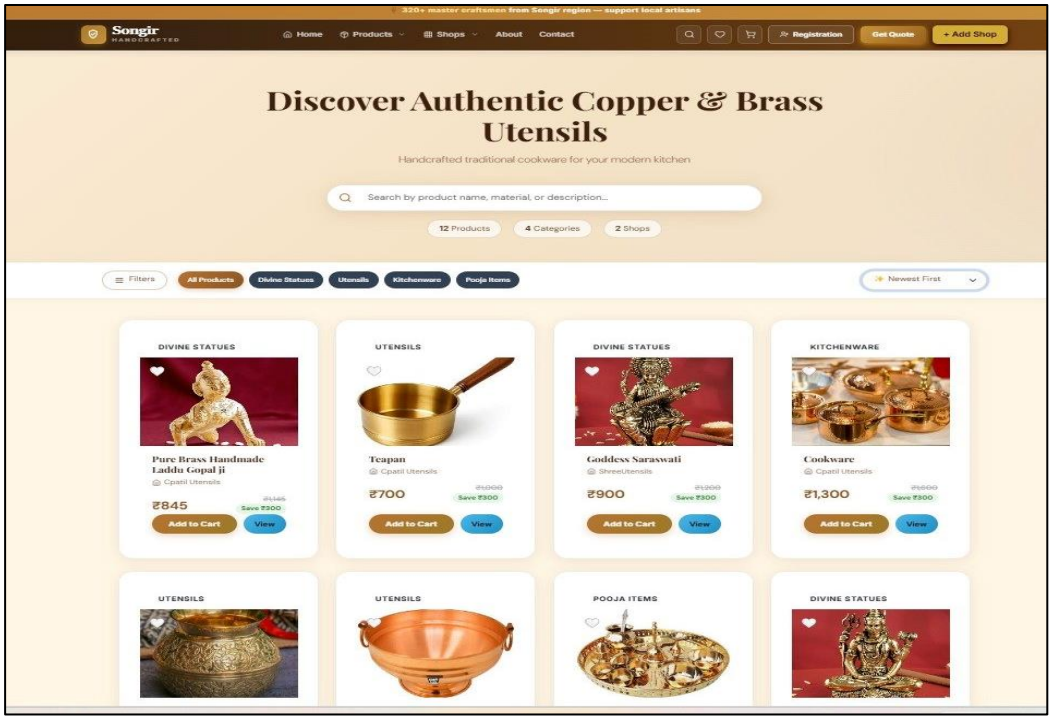

Fig 6.3 Product Page

The Products page is the main catalog showing all handcrafted items available on the platform. It has a search bar at the top, a stats bar, category filter tabs, a sort dropdown, and a grid of product cards. Each card shows category badge, product image, name, shop name, ratings, discounted price, savings badge, and action buttons.

Table 6: Product's DetailsPage Screen Element-wise Explanation

UI Element / Section	Explanation
Page Heading	'Discover Authentic Copper & Brass Utensils' - large centered heading with subtitle 'Handcrafted traditional cookware for your modern kitchen.'
Search Bar	Central search input - customers type product name, material, or description. Results filter in real time.

UI Element / Section	Explanation
Stats Bar	Shows quick counts: 12 Products, 4 Categories, 2 Shops — gives buyers a quick platform overview.
Category Filter Tabs	Pill buttons: All Products, Divine Statues, Utensils, Kitchenware, Pooja Items. Active tab gets highlighted background.
Sort Dropdown	'Newest First' dropdown to sort by date added, price low → high, price high → low, or highest rated.
Product Card — Image	Product photograph with hover zoom effect. Category badge (e.g. DIVINE STATUES, UTENSILS) overlaid at top.
Product Card — Wishlist	Heart icon on top-left of image - click to add/remove from wishlist. Filled = saved, Outline = not saved.
Product Card — Shop Name	Small text below product name showing which artisan shop this product belongs to (e.g. Cpatil Utensils).
Product Card — Price	Shows original MRP struck through (e.g. ₹1,145), current discounted price (e.g. ₹845), and 'Save ₹300' badge in red.
Add to Cart Button	Dark brown button - adds product to cart with quantity = 1. Disabled and greyed if product is Out of Stock.
View Button	Outlined button - navigates to the full Product Detail page for that product.
Out of Stock State	'Out of Stock' label disables the Add to Cart button and shows a greyed card style.

Table 7: Product's DetailsPage Screen APIs Called

Method	Endpoint	Purpose	Body / Params
POST	/api/products	Add new product (with image)	*name, price, category, shopName, shopId, inStock, image*
GET	/api/products	Get all products	-
GET	/api/products/count	Count products for a shop	*?shopName=<n>*
GET	/api/products/compare-list	Product list for compare picker	*?category=<cat>*
GET	/api/products/compare	Compare product across shops	*?name=<productName>*
GET	/api/products/shop/:shopId	Get products of specific shop	*:shopId*
GET	/api/products/:id	Get single product detail	*:id*
PUT	/api/products/:id	Update product info / image	*Fields + image*
DELETE	/api/products/:id	Delete a product	*:id*

Quick Search Modal :

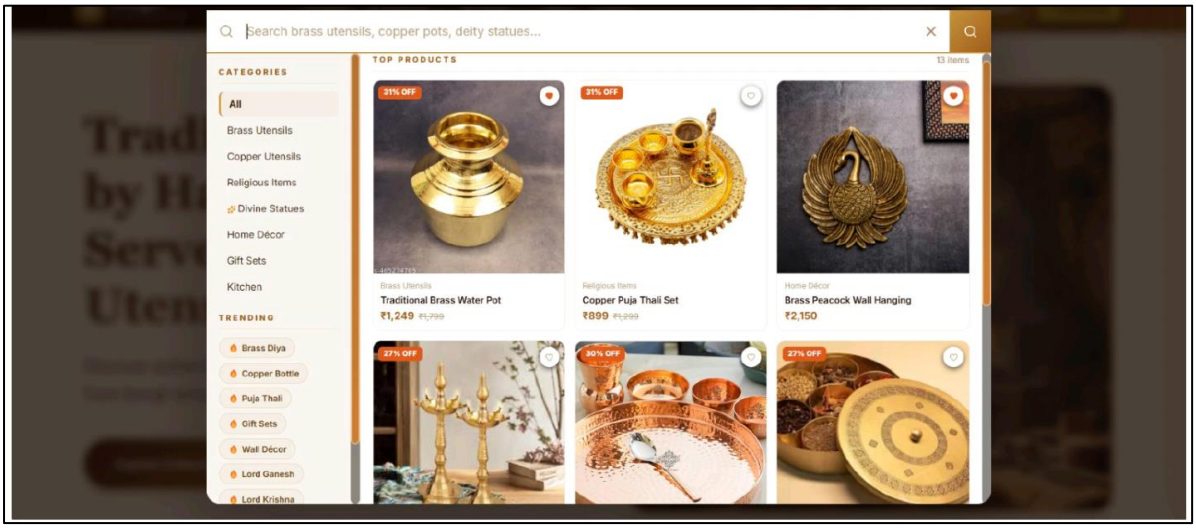

Fig 6.4 Quick Search Modal

The Quick Search Modal appears when the user clicks the Search icon in the navigation bar. It is a full-screen overlay with a blurred background. It provides powerful real-time search experience with categories on the left panel and product results on the right panel

Table 8: Quick Search Modal Screen Element-wise Explanation

UI Element / Section	Explanation
Search Input	Large search box at the top of the modal — user types product name, material, or keyword. Debounced live search activates after 300ms.
Left Panel — CATEGORIES	List of all product categories: All, Brass Utensils, Copper Utensils, Religious Items, Divine Statues, Home Decor, Gift Sets, Kitchen. Click to filter results.
Left Panel — TRENDING	Popular search keyword chips: Brass Diya, Copper Bottle, Puja Thali, Gift Sets, Wall Decor, Lord Ganesh, Lord Krishna. Click to auto-fill search bar.
Right Panel — TOP PRODUCTS	Shows matching product cards with discount badge (27% OFF, 31% OFF, 30% OFF), product image, name, and price.
Results Count	'13 items' count shows total number of matching products found for the current search query.

UI Element / Section	Explanation
Live Search Behavior	Each keystroke triggers a new API request with 300ms debounce to avoid too many requests.
Category Filter Behavior	Clicking a category on the left instantly filters the right panel results to that category only.
Trending Keyword Behavior	Clicking a trending chip auto-fills the search input and triggers a search for that keyword.
Close Modal	Clicking outside the modal or pressing ESC closes the overlay and returns to the previous page.

Table 9 :Quick Search Modal APIs Called

Method	Endpoint	Description	Auth
GET	/api/products/search?q={query}	Real-time product search — returns matching products by name, material, or description with debounce	Public
GET	/api/products?category={cat}	Filter right-panel search results by the selected category from the left panel	Public
GET	/api/products/trending	Fetch trending keywords and popular search terms for the Trending section in the left panel	Public

UI Element / Section	Explanation
Breadcrumb Navigation	'← Back to Products' link at top-left — navigates back to the products listing page.
Value Proposition Cards	3 cards showing key benefits: Multiple Quotes (from top shops), Customization (tailored to needs), Bulk Orders (special pricing).
Form Section 1 — Title	'Your Information' section heading. Collects customer identity details.
Full Name Field	Text input for the customer's full name.
Mobile Number Field	Phone input with +91 India country code prefix for contact number.
Email Address Field	Email input for sending the quote response back to the customer.
Form Section 2 — Title	'Product Details' section heading. Describes what the customer wants.
Select Product Type	Dropdown to choose product category (e.g. Utensils, Divine Statues, Kitchenware).
Select Material	Dropdown to choose material preference (Brass, Copper, Mix, Clay).
Quantity Field	Number input with minimum value of 10 units (for bulk orders).
Select Urgency	Dropdown: Standard (7-10 days), Express (3-5 days), Urgent (1-2 days).
Customization Text Area	Multi-line text area for customer to describe specific customization requirements (e.g. engravings, dimensions, design).
Form Section 3 — Shop	'Select Shop' section — shows registered shops to choose from (e.g. Sharma Brass Works, Patil Copper Crafts).
Submit Button	'Send Quote Request' button — saves quote to DB and sends email notification to selected shop(s).

Custom Quote Request Page

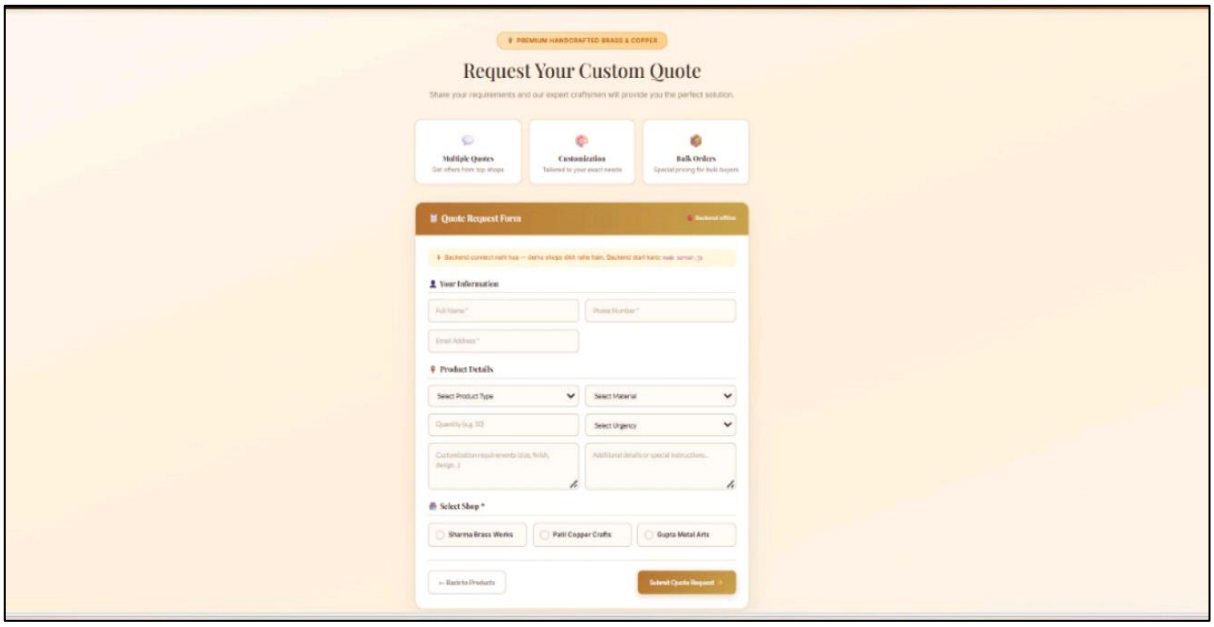

Fig 6.5 Custom Quote Request Page

The Quote Page allows customers to request custom quotes for bulk orders or personalized utensils directly from artisan shops. This is a key B2B feature enabling wholesale inquiries, customization requests, and special orders from multiple shops simultaneously.

Table 10:Custom Quote Request Page Screen Element-wise Explanation

Table 11: Custom Quote Request Page APIs Called

Method	Endpoint	Purpose	Body / Params
POST	/api/quotes	Customer submits quote request	customerName, email, phone, shopId, description, qty
GET	/api/quotes/customer/:email	Customer views their own quotes	:email
GET	/api/quotes/shop/:shopId	Shopkeeper views incoming quotes	:shopId
GET	/api/quotes/:id	Get single quote detail	:id

Method	Endpoint	Purpose	Body / Params
PUT	/api/quotes/:id/respond	Shopkeeper approves / rejects quote	status, responseMessage, quotedPrice
DELETE	/api/quotes/:id	Delete a quote	:id

Register Your Shop Page

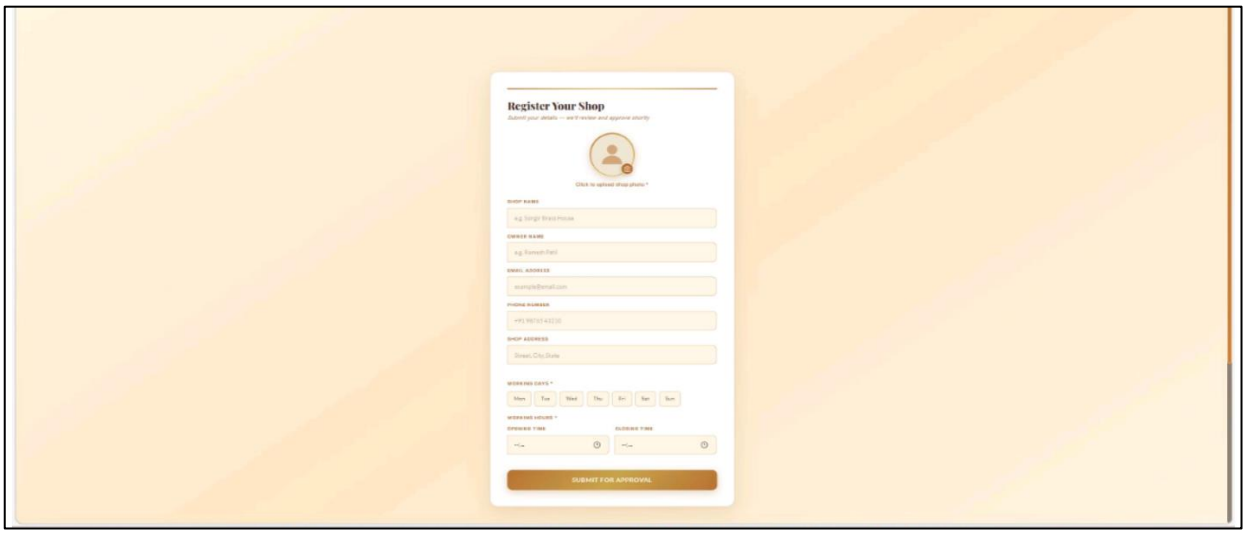

Fig 6.6 Register Your Shop Page

The Add Shop page allows songir artisans/shopkeepers to register their shop on the Songir platform. This form collects all necessary shop details and submits it for Admin approval. Once approved by the Admin, the shop goes live on the website.

Functional Breakdown

Profile Photo Upload: Click to upload shop logo/owner photo .

Shop Name field: e.g. 'My Brass Store'

Owner Name field: e.g. 'Ramesh Mali'

Email Address: Contact email for the shop

Phone Number: With country code (+91 format for India)

Shop Address: Street, City, State

Working Days: Toggle buttons for Mon, Tue, Wed, Thu, Fri, Sat, Sun

Working Hours: Opening Time and Closing Time pickers

SUBMIT FOR APPROVAL: Saves shop data with status='pending', triggers Admin notification

Table 12:Shops (/api/shops)

Method	Endpoint	Purpose	Body / Params
POST	/api/shops	Register new shop	shopName, ownerName, address, phone, workingDays
GET	/api/shops	Get all shops	-
GET	/api/shops/approved	Get only approved shops	-
GET	/api/shops/:id	Get single shop details	:id
PUT	/api/shops/:id	Update shop info	Fields to update
DELETE	/api/shops/:id	Delete a shop	:id

Shopkeeper Dashboard

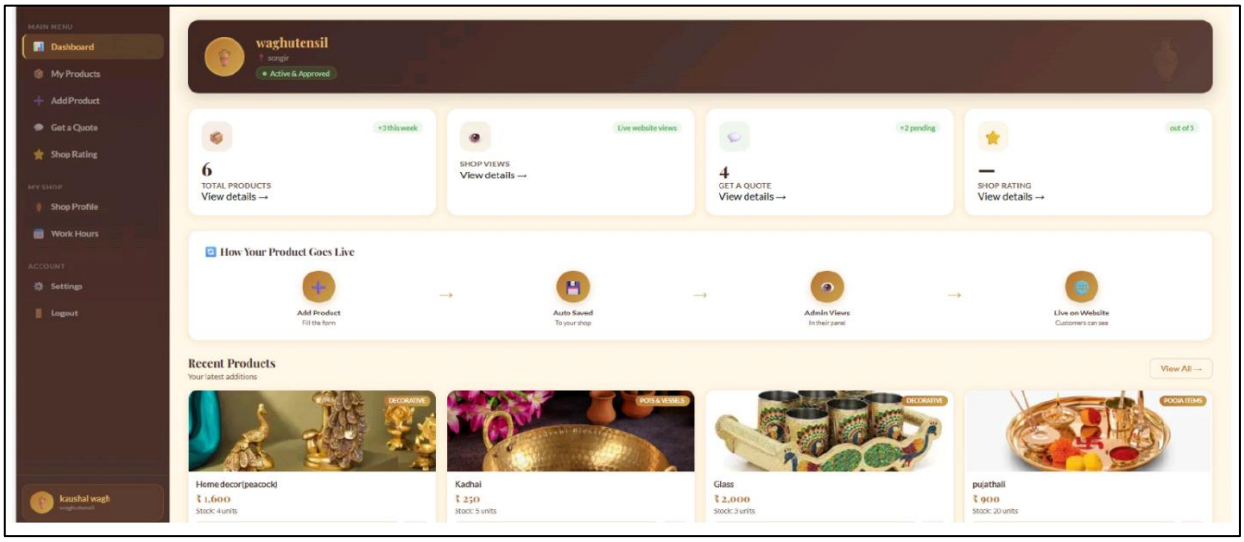

Fig 6.7 Shopkeeper Dashboard

The Shopkeeper Dashboard is the main control panel for artisans after their shop is approved by Admin. It provides a complete overview of shop performance: total products, shop views, pending quote requests, and average star rating. It also shows recent products and a product listing workflow guide.

Table 13: Shopkeeper Dashboard Screen Element-wise Explanation

UI Element / Section	Explanation
Sidebar — Shop Identity	Shows shop name ('Songir Brass') and username ('waghutensil') at the top of the sidebar.
Sidebar — Main Navigation	Dashboard, My Products, Add Product, Get a Quote, Shop Rating — main working pages.
Sidebar — My Shop Section	Shop Profile and Work Hours — sub-pages for managing shop details and operating hours.
Sidebar — Account Section	Settings and Logout options at the bottom of the sidebar.
Stats Card — Total Products	'Total Products: 6' with '+3 this week' label — shows how many products are in the shop.
Stats Card — Shop Views	'Shop Views' counter showing how many customers have visited the shop's page.
Stats Card — Get A Quote	'Get A Quote: 4' with '+2 pending' badge — shows number of pending customer quote requests.
Stats Card — Shop Rating	Average star rating out of 5 based on all customer reviews.
Workflow Section	'How Your Product Goes Live' — 4-step guide: Add Product → Auto Saved → Admin Views → Live on Website.
Recent Products Grid	Shows last 4 added products with image, name, price, and stock count. 'View All' link goes to My Products page.
Welcome Header	Personalized greeting: 'Welcome back, [Owner Name]!' with shop status badge (Approved/Pending).

Table 14 :Shopkeeper Dashboard APIs Called

Method	Endpoint	Description	Auth
GET	/api/shops/my/stats	Fetch dashboard stats: total products count, total shop views, pending quotes count, average rating	Auth
GET	/api/products/my? Limit=4	Fetch the 4 most recently added products for the Recent Products dashboard preview section	Auth
GET	/api/quotes/received	Fetch all pending quote requests sent by customers to this shop	Auth
GET	/api/shops/my/rating	Fetch shop's current calculated average star rating and total review count	Auth

Products Page [Shopkeeper Dashboard's]

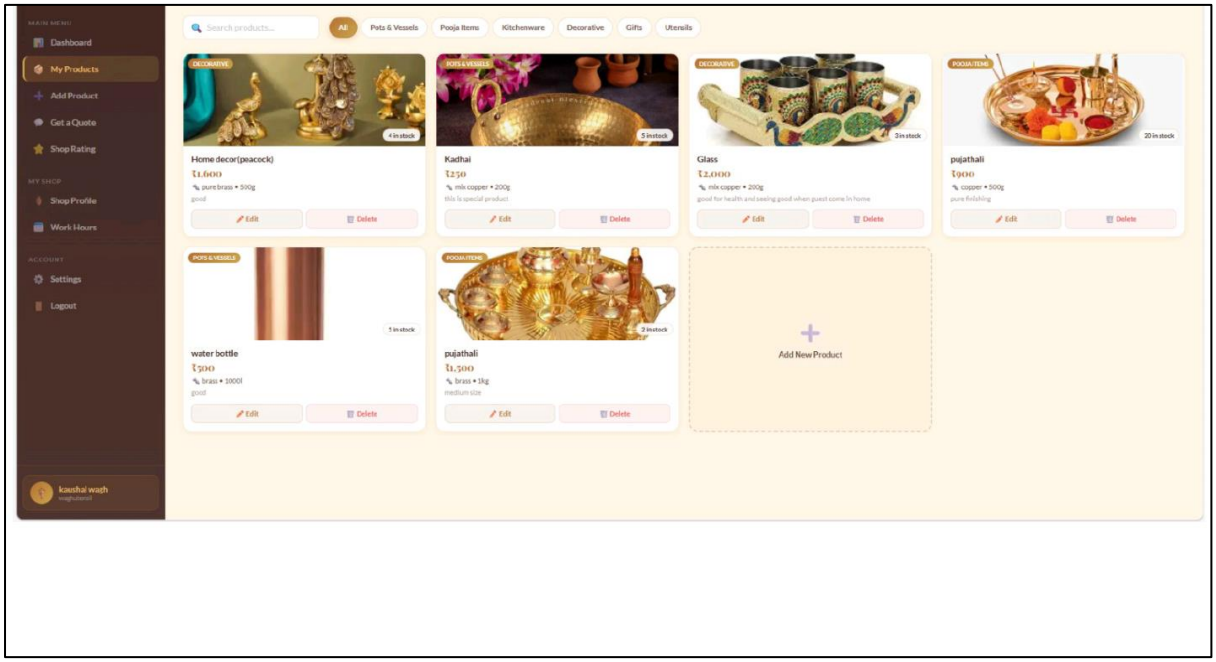

Fig 6.8 Products Page

The My Products page shows all products the shopkeeper has added to their store in a responsive grid view. Each product card has the image, category badge, product name, price, material, weight, description snippet, stock count badge, and action buttons for Edit and Delete. An 'Add New Product' card is also shown.

Table 15: Shopkeeper My Products Page Screen Element-wise Explanation

UI Element / Section	Explanation
Search Bar	Real-time search input at top — filters the product grid by product name as the shopkeeper types.
Category Filter Tabs	Tabs: All, Pots & Vessels, Pooja Items, Kitchenware, Decorative, Gifts, Utensils — filters by category.
Add Product Button	'Add Product' button in top-right corner — navigates to the Add New Product form page.
Product Card — Category Badge	Coloured category pill at top of card (e.g. DECORATIVE, PUJA & POOJA, POOJA ITEMS).
Product Card — Stock Badge	'4 in stock', '5 in stock', '2 in stock' — green badge on image corner showing available quantity.

UI Element / Section	Explanation
Product Card — Product Name	Product name in bold serif text (e.g. Home decor(peacock), Kadhai, Glass, Pujathali).
Product Card — Price	Selling price displayed in large gold text (e.g. Rs. 1,600 / Rs. 250 / Rs. 900 / Rs. 1,500).
Product Card — Material	Material type and weight shown below price (e.g. pure brass 200g, mix copper 200g, copper 550g).
Product Card — Description	Short 2-line description snippet of the product for quick reference.
EDIT Button	Blue 'EDIT' button — opens the product edit form pre-filled with existing product data for update.
DELETE Button	Red 'DELETE' button — shows a confirmation dialog; on confirm, permanently deletes the product.
Add New Product Card	Last card in the grid shows a '+' icon with 'Add New Product' text — shortcut to the Add Product form.

Table 15: Shopkeeper My Products Page APIs Called

Method	Endpoint	Description	Auth
GET	/api/products/my	Fetch all products belonging to the logged-in shopkeeper's shop	Auth
GET	/api/products/my?category={c}	Filter shopkeeper's product grid by the selected category tab	Auth
GET	/api/products/my?search={q}	Real-time search through shopkeeper's own products as they type in the search bar	Auth
PUT	/api/products/:id	Update product details (name, price, stock, description, images) when EDIT form is submitted	Auth

Method	Endpoint	Description	Auth
DELETE	/api/products/:id	Permanently delete a product from the shop after confirmation dialog	Auth

Admin Dashboard

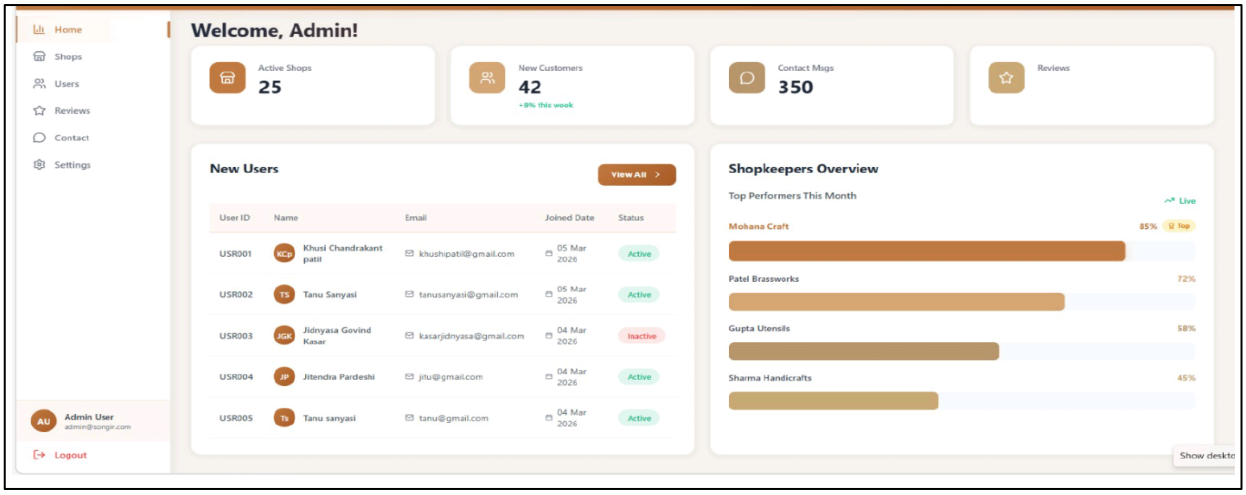

Fig 6.9Admin Dashboard

The Admin Dashboard is the central command center for the platform administrator. It gives a high-level overview of platform health: active shops, new customers, contact messages, and reviews. It also shows a recent user registrations table and a shopkeeper performance leaderboard.

Table 16:Admin Dashboard Screen Element-wise Explanation

UI Element / Section	Explanation
Sidebar — Admin Identity	Shows 'Admin' label with admin email and a logout button at the bottom of the sidebar.
Sidebar Navigation	Home (Dashboard), Shops, Users, Reviews, Contact, Settings — all platform management pages.
Stats Card — Active Shops	'Active Shops: 25' — total number of currently approved and live shops on the platform.

UI Element / Section	Explanation
Stats Card — New Customers	'New Customers: 42' with '+8% this week' indicator — newly registered customers this week.
Stats Card — Contact Msgs	'Contact Msgs: 350' — total contact form submissions received from customers.
Stats Card — Reviews	4th card showing total reviews received across all shops on the platform.
New Users Table	Shows 5 most recently registered users with: User ID (USR001-USR005), avatar initials, full name, email, join date, and Active/Inactive status badge.
View All Users Button	'View All' link button next to the New Users table — navigates to the full Users Management page.
Shopkeepers Overview Panel	Right-side panel showing top performing shops this month with horizontal progress bars and percentage scores.
Performance Leaderboard	Mohana Craft 85% (Top badge), Patel Brassworks 72%, Gupta Utensils 58%, Sharma Handicrafts 43%.
Page Welcome Heading	'Welcome Admin!' personalized greeting at the top of the main content area.

Table 17: Admin Dashboard Screen APIs Called

Method	Endpoint	Description	Auth
GET	/api/admin/stats	Fetch platform stats: total active shops, new customers this week, total messages, total reviews	Admin
GET	/api/users?limit=5&sort=newest	Fetch 5 most recently joined users for the New Users dashboard table	Admin

Method	Endpoint	Description	Auth
GET	/api/admin/shops/performance	Fetch shopkeeper performance percentage scores for the leaderboard right panel	Admin
GET	/api/admin/reviews/recent	Fetch recent reviews for admin awareness panel	Admin

Admin Shop Management

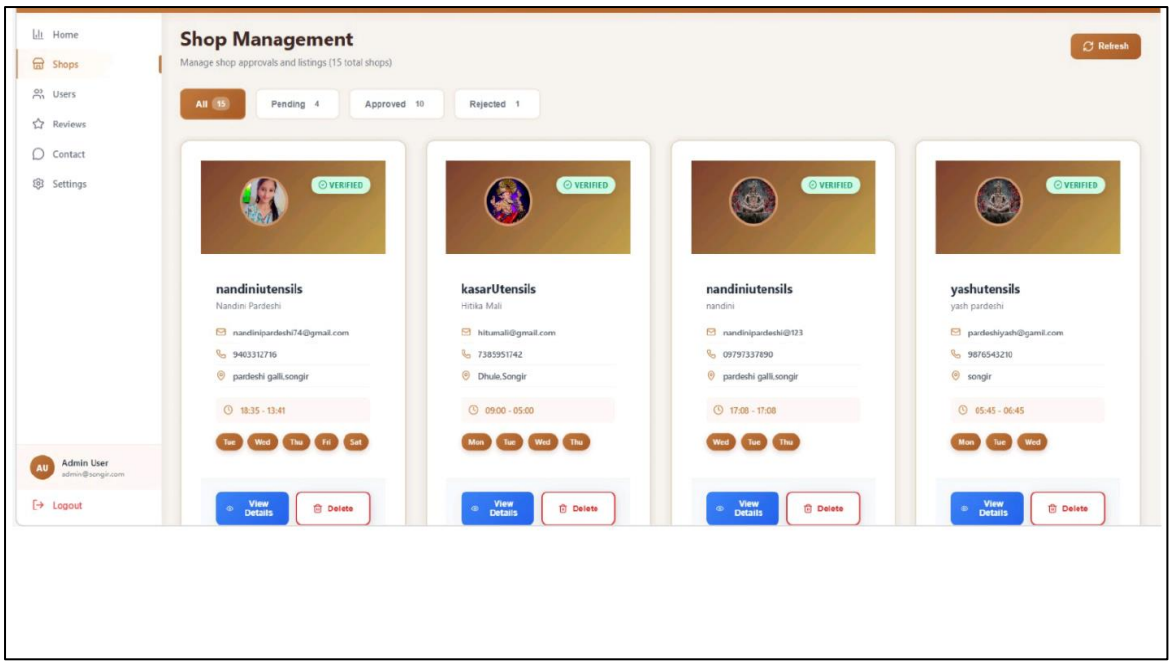

Fig 6.9Admin Shop Management

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Table 18:Admin Shop Management Screen Element-wise Explanation

UI Element / Section	Explanation
Sidebar — Admin Identity	Shows 'Admin' label with admin email and a logout button at the bottom of the sidebar.
Sidebar Navigation	Home (Dashboard), Shops, Users, Reviews, Contact, Settings — all platform management pages.
Stats Card — Active Shops	'Active Shops: 25' — total number of currently approved and live shops on the platform.
Stats Card — New Customers	'New Customers: 42' with '+8% this week' indicator — newly registered customers this week.
Stats Card — Contact Msgs	'Contact Msgs: 350' — total contact form submissions received from customers.
Stats Card — Reviews	4th card showing total reviews received across all shops on the platform.
New Users Table	Shows 5 most recently registered users with: User ID (USR001-USR005), avatar initials, full name, email, join date, and Active/Inactive status badge.
View All Users Button	'View All' link button next to the New Users table — navigates to the full Users Management page.
Shopkeepers Overview Panel	Right-side panel showing top performing shops this month with horizontal progress bars and percentage scores.
Performance Leaderboard	Mohana Craft 85% (Top badge), Patel Brassworks 72%, Gupta Utensils 58%, Sharma Handicrafts 43%.
Page Welcome Heading	'Welcome Admin!' personalized greeting at the top of the main content area.

Table 19:Admin Shop Management APIs Called

Method	Endpoint	Description	Auth
GET	/api/admin/stats	Fetch platform stats: total active shops, new customers this week, total messages, total reviews	Admin
GET	/api/users?limit=5&sort=newest	Fetch 5 most recently joined users for the New Users dashboard table	Admin
GET	/api/admin/shops/performance	Fetch shopkeeper performance percentage scores for the leaderboard right panel	Admin
GET	/api/admin/reviews/recent	Fetch recent reviews for admin awareness panel	Admin

Admin Users Management

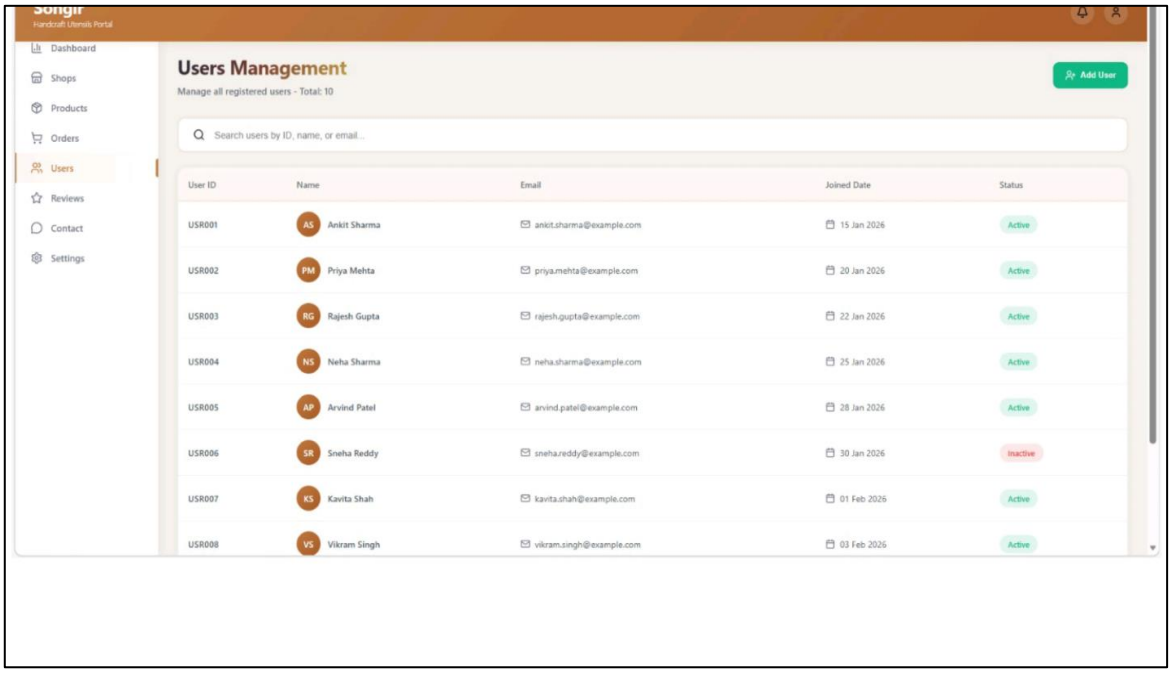

Fig 6.11 Admin Users Management

The Users Management page gives the Admin complete visibility and control over all registered users. It shows a searchable table of all 10 users with their details and status badges, and allows the admin to manually add new user accounts.

Table 20: Admin Users Management Screen Element-wise Explanation

UI Element / Section	Explanation
Page Header	'Users Management — Manage all registered users — Total: 10' — title showing the full user count.
Search Bar	'Search users by ID, name, or email...' — real-time search field that filters the table instantly.
+ Add User Button	Top-right button — opens a modal form to manually create a new user account directly from Admin panel.
Table — User ID Column	Sequential IDs: USR001, USR002... USR008 — unique identifier for each user in the system.
Table — Name Column	Color-coded circular avatar with user initials (e.g. AS, PM, RG) followed by the user's full name.
Table — Email Column	Email icon followed by the user's email address (e.g. ankit.sharma@example.com).
Table — Joined Date	Calendar icon with the date the user registered (e.g. 15 Jan 2026, 20 Jan 2026).
Table — Status Badge	Active (green badge) or Inactive (orange/red badge) — shows current account status.
User Rows (USR001-USR005)	Ankit Sharma, Priya Mehta, Rajesh Gupta, Neha Sharma, Arvind Patel — all Active status.
User Rows (USR006-USR008)	Sneha Reddy (Inactive), Kavita Shah (Active), Vikram Singh (Active).
Toggle Active/Inactive	Admin can click on status badge or a toggle action to switch user between Active and Inactive state.

UI Element / Section	Explanation
Delete User Action	Delete option per user row — permanently removes the user account from the platform.

Table 21: Admin Users Management APIs Called on this Screen

Method	Endpoint	Description	Auth
GET	/api/admin/users	Fetch all registered users with pagination support and full status information	Admin
GET	/api/admin/users?search={q}	Search and filter users by name, email address, or user ID in real time	Admin
POST	/api/admin/users	Manually create a new user account from Admin panel (Add User button)	Admin
PATCH	/api/admin/users/:id/status	Toggle a user's status between Active and Inactive (enable/disable account)	Admin
DELETE	/api/admin/users/:id	Permanently remove a user account and all associated data from the platform	Admin

Database Schema Overview

The Songir portal uses MongoDB (NoSQL) with Mongoose ODM. Below are the main collections and their fields.

Table 22 :Users Collection

Field	Type	Description
_id	ObjectId	MongoDB auto-generated unique document ID
firebaseUid	String	Firebase Authentication UID — primary authentication identifier
Name	String	Full name of the user
Email	String (Unique)	Email address used for login — must be unique across all users
Role	Enum	customer | shopkeeper | admin — determines portal access and permissions
Avatar	String	Cloudinary CDN URL of the user's profile photo
Phone	String	Mobile number with +91 country code
Status	Enum	active | inactive — controlled by admin (inactive users cannot login)
createdAt	Date	Account registration timestamp (auto-set by Mongoose)

Table 23: Shops Collection

Field	Type	Description
_id	ObjectId	Unique shop document ID
ownerId	ObjectId (ref)	Reference to Users collection — the shopkeeper who owns this shop
shopName	String	Official shop name (e.g. waghutensil, kasarUtensils)
ownerName	String	Owner's full name displayed on shop cards

Field	Type	Description
Email	String	Shop contact email for customer inquiries
Phone	String	Shop contact phone number
Address	String	Physical shop address in Songir village
profilePhoto	String	Cloudinary CDN URL for shop banner or logo image
workingDays	Array[String]	Working days array e.g. ['Mon','Tue','Wed','Thu','Fri']
openingTime	String	Shop opening time in 24-hour format (e.g. 09:00)
closingTime	String	Shop closing time in 24-hour format (e.g. 18:00)
Status	Enum	pending | approved | rejected — set by admin through Shop Management panel
Rating	Number	Calculated average star rating (0-5) based on all customer reviews
Views	Number	Total number of times customers have viewed this shop's page
createdAt	Date	Shop registration timestamp

Table 24:Products Collection

Field	Type	Description
_id	ObjectId	Unique product document ID
shopId	ObjectId (ref)	Reference to Shops collection — which shop owns this product
shopName	String	Denormalized shop name for fast display without joins
Name	String	Product name (e.g. Traditional Brass Water Pot)

Field	Type	Description
category	String	Category: Decorative, Kitchenware, Pooja Items, Utensils, Divine Statues
material	Enum	brass | copper | mix | clay — raw material used
Weight	String	Product weight (e.g. 200g, 1kg, 550g)
Price	Number	Selling price in INR (Rs.)
originalPrice	Number	MRP before discount (shown as struck-through price)
Stock	Number	Available quantity in stock
description	String	Detailed product description text
Images	Array[String]	Array of Cloudinary CDN image URLs for the product gallery
Badge	String	Display badge: POPULAR | BEST SELLER | PREMIUM | NEW
Rating	Number	Average product rating (0-5) from customer reviews
inStock	Boolean	true/false — controls whether Add to Cart is enabled
isActive	Boolean	Whether product is currently visible to customers
createdAt	Date	Product listing creation timestamp

Table 25:Technologies Mastered

Area	**Technologies Used**
Frontend	React.js 18+, React Router DOM, Context API, Axios, CSS, Tailwind CSS
Backend	Node.js, Express.js, REST API design, Middleware, CORS, JWT Authentication
Database	MongoDB Atlas, Mongoose ODM, Schema design, Indexing, Aggregation Pipelines
Cloud Services	Firebase Authentication (JWT), Cloudinary (image upload and CDN delivery)

CHAPTER 7

TESTING AND EVALUATION

7.1 INTRODUCTION

This chapter covers the testing activities performed for the Songir Handcraft Utensil Portal — a web-based e-commerce connecting traditional brass and copper utensils artisan shopkeepers in Songir village with customers across India.

Testing is a critical phase of the software development lifecycle that ensures the system meets its specified requirements and functions correctly in all scenarios. The objective of this testing phase is to:

- Verify that all authentication and role-based access mechanisms work correctly for Customers, Shopkeepers, and Admin.
- Ensure that Shop Registration, Admin Approval, and Product Management modules function as specified in the SRS.
- Validate the core features: Product Comparison (up to 4 products) and Get a Quote module with email notifications.
- Confirm that Review & Rating and Admin Dashboard modules operate without defects.
- Identify and document any defects for timely resolution before final release.

Testing was structured to cover both positive scenarios (valid inputs, successful operations) and negative scenarios (invalid inputs, unauthorized access, boundary conditions).

Testing was performed using the following tools:

- Selenium IDE: Record-and-playback UI test automation tool for quick scenario capture and regression checks.
- Selenium WebDriver: Scripted browser automation tool for repeatable, assertion-based test execution.
- Postman: API-level testing for backend REST endpoints.

Modules covered in this chapter:

- User Authentication Module
- Shop Management Module
- Product Management Module
- Product Comparison Module (Core Feature)
- Get a Quote Module (Core Feature)

- Review & Rating Module
- Admin Dashboard Module
- Shopkeeper Dashboard Module

Out of scope for this chapter:

- Payment gateway testing (not implemented in this version).
- Real-time product tracking testing (not in scope).

7.2 TEST PLAN

The test plan defines the objectives, scope, testing environment, tools, entry and exit criteria, and risk factors associated with the testing cycle of the Songir Handcraft Utensil Portal. This test plan has been prepared following IEEE standard format to ensure a well-structured and standardized documentation approach.

Test Plan ID	TP_SONGIR_01
Project Name	Songir Handcraft Utensils Portal
Version	1.0
Date	April 2026

1. Introduction

This test plan verifies that all modules of the Songir Handcraft Utensil Portal work correctly as specified in the SRS. It ensures valid users can perform their roles successfully, invalid inputs are handled gracefully, and the core features - Product Comparison and Get a Quote - function reliably. It also validates security mechanisms including JWT-based authentication, role-based access control, and bcrypt password hashing.

2. Test Items

- Login form UI - input fields, submit button, error messages, role-based routing
- Shop registration form and admin approval/rejection workflow
- Product add/edit/delete with image upload and stock management
- Product Comparison module - add up to 4 products, side-by-side view, remove
- Get a Quote module - form submission, shopkeeper response, email notification

- Review & Rating - submit, view, helpful vote, shopkeeper delete
- Admin Dashboard - user/shop/product/quote/review management

3. Features to be Tested

- User Authentication: Register, Login, Logout, JWT session management
- Role-based access control (Customer, Shopkeeper, Admin)
- Shop Registration and Admin Approval workflow
- Product CRUD operations with image upload
- Product Filtering by category and shop name
- Product Comparison (core feature)
- Quote Request submission, shopkeeper response, email notification (core feature)
- Review and Rating system with helpful votes
- Admin Dashboard operations

4. Features Not to be Tested

- Direct online payment gateway (not implemented in this version)
- Real-time courier/shipment tracking
- Cloud storage (AWS S3) for images
- API load testing and performance benchmarking under 50+ concurrent users

5. Approach

- Functional and non-functional testing will be performed across all modules
- Positive and negative test cases will be executed for each module
- Manual testing, Selenium IDE, Selenium WebDriver, and Postman will be used
- UI validation testing will be performed for all forms
- Role-based access control will be tested for all three user roles

6. Pass / Fail Criteria

- All high-priority test cases must pass successfully
- Minimum 95%–99% requirement coverage should be achieved
- System should not have any critical or blocking defects at release
- If blocking defects are found, testing will be paused and resumed after fixing
- Expected results must match actual results for all critical scenarios
- All major defects must be fixed and retested before project completion

7. Suspension Criteria and Resumption Requirements

- Testing will be suspended if critical bugs preventing core feature usage are found
- Testing will resume after bug fixing, developer sign-off, and system stabilization

8. Test Deliverables

- Test plan document
- Test cases document (for all 7 modules)
- Defect report (per defect found)
- Test summary report

9. Test Tasks

- Prepare test plan and test cases for all modules
- Execute test cases in the test environment
- Identify, log, and track defects
- Retest fixed defects and update defect status
- Prepare final test summary report

10. Test Environment

- Operating System: Windows 11
- Browser: Google Chrome (v90+), Microsoft Edge (v90+)
- Frontend: React.js + Tailwind CSS (Vite, localhost)
- Backend: Node.js + Express.js
- Database: MongoDB (Mongoose ODM)
- Authentication: JWT (JSON Web Tokens)
- Email Service: Nodemailer (Gmail SMTP)
- File Storage: Multer (Local Disk /uploads)
- Server: Localhost / Test Server
- Tools: Selenium IDE, Selenium WebDriver, Postman
- Network: Active internet connection required

11. Responsibilities

- QA Lead: Review test artifacts and approve test plan
- Test Engineer: Execute test cases, log defects, and prepare reports

- Developer: Fix reported defects and perform unit testing
- Project Manager: Approve release decision based on test summary

12. Staffing and Training

Requires knowledge of portal functionality across all three user roles, web testing techniques, Selenium IDE/WebDriver, and Postman API testing. Familiarity with JWT authentication and MongoDB is beneficial.

13. Scheduling

Project Deadline: April 2026

14. Risks

- Incomplete test data or missing test accounts for all three user roles
- Gmail SMTP dependency for email notification testing
- Browser compatibility issues on different Chrome/Edge versions
- Changing product requirements during the testing cycle
- Multer file upload errors affecting product image test cases

15. Approvals

Project Guide, Team Members

7.2.1 Test Cases :- User Login

Table 26 : User Login Module

Test Case ID	Objective	Input / Steps	Expected Result	Actual Result	Status
TC_LOGIN_001	Valid Admin Login	Email: admin@songir.com Password: admin123	Admin redirected to Admin Dashboard	Admin logged in & redirected to Dashboard	Pass
TC_LOGIN_002	Valid Customer Login	Email: customer@gmail.com Password: pass123	Customer redirected to Home Page	Customer logged in & redirected to Home	Pass

Test Case ID	Objective	Input / Steps	Expected Result	Actual Result	Status
TC_LOGIN_003	Valid Shopkeeper Login	Email: artisan@songir.com Password: artisan123	Shopkeeper redirected to Shopkeeper Dashboard	Shopkeeper logged in successfully	Pass
TC_LOGIN_004	Invalid Password	Email: customer@gmail.com Password: wrongpass	Show 'Invalid credentials' error	Error message displayed, login blocked	Pass
TC_LOGIN_005	Invalid Email	Email: notexist@gmail.com Password: anypass	Show 'User not found' error	Error message displayed	Pass
TC_LOGIN_006	Empty Email Field	Email: blank Password: pass123	Show 'Email is required'	Validation message displayed	Pass
TC_LOGIN_007	Empty Password Field	Email: customer@gmail.com Password: blank	Show 'Password is required'	Validation message displayed	Pass
TC_LOGIN_008	Both Fields Empty	Email: blank Password: blank	Show 'Please fill in all fields'	Validation message displayed	Pass
TC_LOGIN_009	Invalid Email Format	Email: customeratgmail Password: pass123	Show 'Invalid email format'	Validation message displayed	Pass
TC_LOGIN_010	Role-Based Redirection	Login as customer & check URL	Only redirected to Customer Home, not Admin	Redirected to correct home page	Pass

Test Case ID	Objective	Input / Steps	Expected Result	Actual Result	Status
TC_LOGIN_011	Forgot Password	Click Forgot Password, enter email	Password reset link sent to email	Reset link not sent, feature not working	Fail
TC_LOGIN_012	Multiple Failed Attempts	Enter wrong password 5+ times	Account locked / warning shown	System allows unlimited attempts, no restriction	Fail

7.2.2 Test Cases :- Shop Management Module

Table 27 : Shop Management Module

Test Case ID	Objective	Input / Steps	Expected Result	Actual Result	Status
TC_SHOP_001	Shop Registration by Shopkeeper	Fill all shop fields and submit	Shop created with 'pending' status	Shop created successfully with pending status	Pass
TC_SHOP_002	Admin Approves Shop	Admin views pending shops, clicks Approve	Shop status changes to 'approved'	Shop approved and visible to customers	Pass
TC_SHOP_003	Admin Rejects Shop	Admin clicks Reject, adds reason	Shop status changes to 'rejected' with reason	Rejection reason saved correctly	Pass
TC_SHOP_004	Approved Shop Visible to Customer	Customer visits Shop Directory	Only approved shops are listed	Approved shops displayed correctly	Pass

Test Case ID	Objective	Input / Steps	Expected Result	Actual Result	Status
TC_SHOP_005	Shop Registration with Missing Fields	Submit form without Shop Name	Show field validation error	Validation error displayed	Pass
TC_SHOP_006	Shopkeeper Cannot Add Products Before Approval	Pending shopkeeper tries to add product	System blocks product addition	Access denied, product form not accessible	Pass

7.2.3 Test Cases :- Product Management Module

Table 28 : Product Management Module

Test Case ID	Objective	Input / Steps	Expected Result	Actual Result	Status
TC_PROD_001	Add New Product	Shopkeeper fills product form (name, price, material etc.) and submits	Product listed in portal with correct details	Product added successfully	Pass
TC_PROD_002	Edit Existing Product	Shopkeeper edits product price and description	Product updated with new values	Product updated correctly	Pass
TC_PROD_003	Delete Product	Shopkeeper clicks delete on a product	Product removed from listing and image deleted	Product deleted successfully	Pass
TC_PROD_004	Product Image Upload	Upload image < 5MB during add product	Image uploaded within 5 seconds	Image uploaded and saved	Pass

Test Case ID	Objective	Input / Steps	Expected Result	Actual Result	Status
TC_PROD_005	Stock Qty Set to 0	Update stockQty to 0	Product marked as 'Out of Stock'	Out of Stock flag updated automatically	Pass
TC_PROD_006	Filter Products by Category	Customer selects 'Brass' category filter	Only Brass products displayed	Filter working correctly	Pass
TC_PROD_007	Filter Products by Shop Name	Customer selects specific shop in filter	Only that shop's products displayed	Shop filter working correctly	Pass
TC_PROD_008	View Product Detail Page	Click on any product card	Detailed product page opens with all attributes	Product detail page displayed correctly	Pass
TC_PROD_009	Add Product Without Required Fields	Submit product form without Category	Show validation error	Validation error displayed	Pass

7.2.4 Test Cases :- Product Comparison Module(Core features)

Table 29 : Product Comparison Module

Test Case ID	Objective	Input / Steps	Expected Result	Actual Result	Status
TC_COMP_001	Add Product to Comparison	Click 'Add to Compare' on product card	Product added to comparison list	Product added successfully	Pass
TC_COMP_002	Add Same Product Twice	Click 'Add to Compare' on same product again	System prevents duplicate addition	Duplicate blocked with message	Pass

Test Case ID	Objective	Input / Steps	Expected Result	Actual Result	Status
TC_COMP_003	Add More Than 4 Products	Try adding 5th product to comparison	System shows 'Max 4 products' error	Max limit enforced correctly	Pass
TC_COMP_004	View Comparison Page	Click 'Compare' button after adding products	Side-by-side table with all attributes shown	Comparison page rendered correctly	Pass
TC_COMP_005	Remove Product from Comparison	Click remove on one product in comparison	That product removed from table	Product removed successfully	Pass
TC_COMP_006	Comparison Resets on Logout	Login, add to compare, logout	Comparison list clears after logout	Comparison list reset correctly	Pass
TC_COMP_007	Attribute Differences Highlighted	Compare two products with different prices	Price difference visually highlighted	Highlighting not implemented	Fail

7.2.5 Test Cases — Get a Quote Module (Core Feature)

Table 30 : Get a Quote Module

Test Case ID	Objective	Input / Steps	Expected Result	Actual Result	Status
TC_QUOTE_001	Submit Quote as Logged-in Customer	Fill quote form (name, phone, qty, budget) and submit	Quote saved with 'pending' status	Quote submitted successfully	Pass
TC_QUOTE_002	Submit Quote Without Login	Guest user clicks 'Get a Quote'	Redirect to login page	Redirect to login working	Pass

Test Case ID	Objective	Input / Steps	Expected Result	Actual Result	Status
TC_QUOTE_003	Quote Form Missing Required Field	Submit without Phone Number	Show field validation error	Validation error displayed	Pass
TC_QUOTE_004	Shopkeeper Views Quotes	Shopkeeper opens quote dashboard	All quotes for their shop displayed	Quotes listed correctly	Pass
TC_QUOTE_005	Shopkeeper Responds to Quote	Shopkeeper sets status to 'Quoted', adds note and price	Quote status updated, email sent to customer	Status updated, email notification triggered	Pass
TC_QUOTE_006	Customer Receives Email Notification	Shopkeeper responds to customer quote	HTML email with status, note, price received by customer	Email received with correct content	Pass
TC_QUOTE_007	Customer Views Quote History	Customer opens dashboard > My Quotes	List of all past quote requests with status shown	Quote history displayed correctly	Pass
TC_QUOTE_008	Filter Quotes by Status	Shopkeeper filters by 'Pending'	Only pending quotes shown	Status filter working correctly	Pass

7.2.6 Test Cases - Review & Rating Module

Table 31 : Review & Rating Module

Test Case ID	Objective	Input / Steps	Expected Result	Actual Result	Status
TC_REV_001	Submit Review as Customer	Fill review (rating, title, body) and submit	Review saved and shown on product page	Review submitted successfully	Pass
TC_REV_002	Submit Two Reviews Same Product	Same customer tries to review same product again	System blocks second review	Duplicate review blocked	Pass
TC_REV_003	Guest Views Reviews	Non-logged-in user opens product page	All reviews visible without login	Reviews visible to guests	Pass
TC_REV_004	Mark Review as Helpful	Customer clicks 'Helpful' on a review	Helpful counter increments by 1	Counter incremented correctly	Pass
TC_REV_005	Shopkeeper Deletes Inappropriate Review	Shopkeeper clicks delete on a review	Review removed from product page	Review deleted successfully	Pass
TC_REV_006	Average Rating Update After Review	Customer submits 5-star review on a 3-star product	Average rating recalculated and updated	Rating updated correctly	Pass

7.2.7 Test Cases - Admin Dashboard Module

Table 32 : Review & Rating Module

Test Case ID	Objective	Input / Steps	Expected Result	Actual Result	Status
TC_ADMIN_001	Admin Views All Users	Admin opens User Management page	All registered users listed with status and timestamps	Users listed correctly	Pass
TC_ADMIN_002	Admin Views All Shops	Admin opens Shop Management page	All shops (pending/approved /rejected) listed	Shops listed with correct status	Pass
TC_ADMIN_003	Admin Views All Products	Admin opens Product Overview page	All products across all shops displayed	Products listed correctly	Pass
TC_ADMIN_004	Admin Views All Quotes	Admin opens Quote Overview page	All quote requests with statuses displayed	Quotes displayed correctly	Pass
TC_ADMIN_005	Admin Deletes Inappropriate Review	Admin views review list, clicks delete	Review permanently removed	Review deleted successfully	Pass
TC_ADMIN_006	Admin Deletes a Shop	Admin clicks delete on a shop	Shop removed from portal	Shop deleted successfully	Pass

7.2.8 Test Cases For Registration (User) :

Table 33: Test Cases For Registration (User)

Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output
TC_USU_01	Valid Registration	Enter valid name, email, password, phone → Click Sign Up	Account created successfully	Account created successfully
TC_USU_02	Empty Fields Validation	Leave all fields blank → Click Sign Up	Error: All fields are required	Error message displayed
TC_USU_03	Invalid Email Format	Enter email without '@' (e.g. abc.com)	Error: Invalid email format	Error shown
TC_USU_04	Weak Password	Enter password below required length	Error: Password too short	Error shown
TC_USU_05	Duplicate Email	Enter an already registered email	Error: Email already exists	Error shown
TC_USU_06	Invalid Phone Number	Enter fewer than 10 digits	Error: Invalid phone number	Error shown
TC_USU_07	Password Mismatch	Enter different password and confirm password	Error: Passwords do not match	Error shown
TC_USU_08	Successful Redirect	Complete valid registration	Redirect to dashboard	Redirect successful
TC_USU_09	Special Characters in Name	Enter name with special characters (e.g. @#%)	Error: Invalid name format	Error shown
TC_USU_10	Numeric Value in Name	Enter numbers in name field (e.g. 12345)	Error: Name should contain only letters	Error shown
TC_USU_11	Very Long Input Values	Enter very long name or email	System restricts input or shows error	Input restricted

TC_USU_12	SQL Injection Check	Enter SQL query (' OR 1=1) in fields	System should reject input securely	Input rejected
TC_USU_13	Trim Spaces Handling	Enter spaces before/after email or name	System trims spaces and processes correctly	Works correctly
TC_USU_14	Network Failure During Sign-Up	Turn off internet and submit form	Error: Network error / Try again	Error shown
TC_USU_15	Multiple Clicks on Sign Up	Click Sign Up button multiple times quickly	Only one account should be created	Single account created

7.3 DEFECT REPORTS

The defect report is a crucial document used to record and track defects identified during the testing process of the Songir Handcraft Utensil Portal. It provides detailed information about each defect, including its description, severity, priority, steps to reproduce, and current status.

This report helps the development and testing teams to understand, prioritize, and resolve issues efficiently. It ensures effective communication among team members and maintains transparency throughout the defect management process.

Defect Report 1

Table 33 : Defect Report 1

Defect ID	DR_SONGIR_001
Project Name	Songir Handcraft Utensil Portal
Product	Web Application
Release Version	v1.0
Module	User Authentication Module — Forgot Password
Detected Build Version	v1.1

Summary	Password reset link is not sent to the registered email address.
Description	While executing the login module test scenarios, it was observed that the 'Forgot Password' functionality is not working properly. When the user enters a valid registered email address and clicks the submit button, the system does not send any password reset link to the email. The application also does not display any success or error message. Due to this, the user is unable to reset the password and regain account access.
Steps to Reproduce	1. Open the login page 2. Click on 'Forgot Password' link 3. Enter a valid registered email address 4. Click on Submit / Reset button 5. Observe: No email received, no message shown
Expected Result	System should send a password reset link to the registered email and allow user to reset their password.
Actual Result	No reset link is sent. No success or error message is displayed.
Severity	High (User is unable to reset password and cannot regain account access)
Priority	High (Affects real users, needs immediate fix)
Reported By	Test Engineer
Status	Open

Defect Report 2

Table 34 : Defect Report 2

Defect ID	DR_SONGIR_002
Project Name	Songir Handcraft Utensil Portal
Product	Web Application
Release Version	v1.0
Module	Product Comparison Module — Attribute Highlighting
Detected Build Version	v1.1
Summary	Differences between product attributes are not visually highlighted on the Comparison Page.
Description	While testing the Product Comparison module (core feature), it was observed that when two or more products with different attribute values (e.g., different prices, materials, or stock status) are displayed in the comparison table, the differing values are not visually highlighted or differentiated. As per the SRS (FR-COMP-02 and NFR 4.3), the Comparison Page shall visually highlight differences between product attributes. Currently, all values appear in uniform styling, making it difficult for customers to identify differences at a glance.
Steps to Reproduce	1. Login as a customer 2. Add 2 products with different prices and materials to comparison 3. Navigate to the Comparison Page 4. Observe: All attribute values are displayed in the same style without any visual highlighting of differences

Expected Result	Attributes that differ between products should be visually highlighted (e.g., colored background, bold text, or border) for easy comparison.
Actual Result	All attribute values are displayed uniformly. No visual differentiation is applied for differing values.
Severity	Medium (Core comparison feature works, but usability is reduced)
Priority	Medium (Important for user experience — to be fixed before release)
Reported By	Test Engineer
Status	Open

Defect Report 3

Table 35 : Defect Report 3

Defect ID	DR_SONGIR_003
Project Name	Songir Handcraft Utensil Portal
Product	Web Application
Release Version	v1.0
Module	User Authentication Module — Security (Failed Login Attempts)
Detected Build Version	v1.1
Summary	No restriction or warning is applied after multiple failed login attempts.

Description	While testing the login module, it was observed that the system allows unlimited login attempts even after entering incorrect passwords multiple times. The application does not display any warning message or implement any account lockout mechanism. This creates a critical security vulnerability, as malicious users or automated bots can perform brute-force attacks to guess user passwords and gain unauthorized access to customer, shopkeeper, or admin accounts.
Steps to Reproduce	1. Open the login page 2. Enter a valid registered email address 3. Enter an incorrect password 4. Click the Login button 5. Repeat steps 3–4 more than 5 times 6. Observe: No warning, no lockout, login page remains fully accessible
Expected Result	After 3–5 failed login attempts, the system should temporarily block the account or display a security warning message.
Actual Result	The system allows unlimited login attempts without any restriction, warning, or lockout mechanism.
Severity	High (Security vulnerability — risk of brute-force attack and unauthorized access)
Priority	High (Needs immediate fix to protect all user accounts)
Reported By	Test Engineer
Status	Open

7.4 TEST SUMMARY REPORT

This section provides a consolidated summary of all test cases executed across all modules of the Songir Handcraft Utensil Portal, along with the overall quality assessment of the system.

Table 36 : Test Summary Report

Module	Total TCs	Passed	Failed	Pass %
User Authentication Module	12	10	2	83%
Shop Management Module	6	6	0	100%
Product Management Module	9	9	0	100%
Product Comparison Module	7	6	1	86%
Get a Quote Module	8	8	0	100%
Review & Rating Module	6	6	0	100%
Admin Dashboard Module	6	6	0	100%
TOTAL	**54**	**51**	**3**	**94.4%**

Overall Quality Assessment

A total of 54 test cases were executed across 7 modules of the Songir Handcraft Utensil Portal. 51 test cases passed and 3 test cases failed, resulting in an overall pass rate of 94.4%. The 3 failures are associated with 3 open defects:

- DR_SONGIR_001: Forgot Password functionality not working (High severity — Open)
- DR_SONGIR_002: Product Comparison attribute highlighting not implemented (Medium severity — Open)
- DR_SONGIR_003: No restriction on multiple failed login attempts — brute-force vulnerability (High severity — Open)

The core modules — Shop Management, Product Management, Get a Quote, Review & Rating, and Admin Dashboard — achieved 100% pass rate. The system is functionally stable for primary

workflows. The 3 identified defects must be resolved before final deployment to ensure security and completeness of the core comparison feature.

Testing Phase:

The testing phase is an important part of the system development process. After the development of the Songir Handcraft Utensil Portal, the system was thoroughly tested to ensure that all features and functionalities work correctly and efficiently. The main objective of testing was to identify and remove errors, ensure smooth system performance, and verify that the system meets user requirements.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

CONCLUSION:

The Songir Handcrafted Brass & Copper Utensils Online Marketplace is developed to support the traditional utensil artisans of Songir by providing a digital platform to showcase their shops and products. Through this system, customers can explore different utensil shops, view handcrafted products, and send quote requests according to their requirements. This helps customers easily discover authentic utensils and connect with multiple shopkeepers in one place. The platform also includes an "Add My Shop" functionality, which allows utensil artisans or shopkeepers from Songir to register their shops and display their handcrafted utensils online.

This feature helps local businesses increase their visibility and reach more customers beyond their physical location

The future, the system can be improved by adding features such as online payment integration, order tracking. The platform can also be expanded to include more artisans, improved product catalogs, and better communication tools between customers and shopkeepers. These improvements will further strengthen the digital presence of Songir utensil artisans and help promote traditional craftsmanship on a larger scale.

FUTURE SCOPE:

Mobile Application Development

In the future, a mobile application can be developed to allow users to access the Songir Handcraft Utensils Portal easily through smartphones and tablets.

Online Payment Integration

Secure payment gateways such as UPI, debit card, credit card, and net banking can be integrated to provide safe and convenient online transactions.

Real-Time Order Tracking

A real-time tracking system can be implemented so customers can monitor the status of their orders from dispatch to delivery.

Advanced Search and Filtering

Smart search options and filters can be implemented to help users quickly find specific handcrafted utensils based on category, price, or popularity.

CHAPTER 9

REFERENCE AND BIBLIOGRAPHY

REFERENCE :

WEB & ONLINE SOURCES

1. Thinkitchen. (n.d.). Website interface and user experience design inspiration. Retrieved from https://www.thinkitchen.in/

   Used as UI/UX design reference for the e-commerce product listing and shop layout.
2. Scribd. (n.d.). Final documentation on metal craft. Retrieved from https://www.scribd.com/document/276015551/01-Final-Documentation-on-Metal-Craft

   Used as background reference for traditional METTAL craft history and production methods.

TECHNOLOGY & FRAMEWORK DOCUMENTATION

1. Meta Platforms, Inc. (2024). React – A JavaScript library for building user interfaces. Retrieved from https://react.dev/

   Official documentation used for React.js frontend development (components, hooks, routing).
2. MongoDB, Inc. (2024). MongoDB manual – NoSQL database documentation. Retrieved from https://www.mongodb.com/docs/
3. npm, Inc. (2024). Express.js – Fast, unopinionated, minimalist web framework for Node.js. Retrieved from https://expressjs.com/

   Express.js used as the backend REST API framework for all API routes and middleware.
4. Axios Contributors. (2024). Axios – Promise based HTTP client for the browser and Node.js. Retrieved from https://axios-http.com/docs/intro

   Used in the React frontend for all HTTP requests to the backend API.

ACADEMIC & RESEARCH REFERENCES

1. Government of India, Ministry of Textiles. (2022). Annual report on handicrafts and traditional crafts of India. Ministry of Textiles, Government of India.

   Used to support background context on India's handicraft industry and artisan communities. MSME Development Institute. (2021). Brass and copper utensil manufacturing – Cluster development report, Nashik. Government of India, MSME.

   Referenced for understanding the brass/copper craft ecosystem in the Nashik district region including Songir.

BIBLIOGRAPHY:

PROJECT & DESIGN INSPIRATION

1. Thinkitchen. (n.d.). Website interface and user experience design inspiration. Retrieved from https://www.thinkitchen.in/
2. Meesho. (n.d.). Meesho – Online shopping app for fashion, home & kitchen. Retrieved from https://www.meesho.com/

   Studied as a reference for product card layout, wishlist UX, and mobile-first e-commerce design patterns.
3. Flipkart Internet Pvt. Ltd. (n.d.). Flipkart – India's e-commerce marketplace. Retrieved from https://www.flipkart.com/

   We Studied for cart system, product detail page structure, and recently-viewed feature implementation.

METAL CRAFT & ARTISAN RESEARCH

1. Scribd. (n.d.). Final documentation on metal craft. Retrieved from https://www.scribd.com/document/276015551/01-Final-Documentation-on-Metal-Craft
2. React Router Contributors. (2024). React Router – Declarative routing for React. Retrieved from https://reactrouter.com/en/main

Used for client-side routing across all pages (Home, Products, Shops, Quote, Dashboard, Admin).

3. Multer Contributors. (2024). Multer – Node.js middleware for handling multipart/form-data. Retrieved from https://github.com/expressjs/multer

Used for product image and shop photo file upload handling on the backend.

4. [B17] npm, Inc. (2024). Styled-components – Visual primitives for the component age. Retrieved from https://styled-components.com/docs

Used for CSS-in-JS component styling with theming support across the React frontend.